



Design Narrative
23-0033 MRSG DST STORAGE FACILITY
MARSOC
CAMP LEJEUNE, Jacksonville, NC

CONTRACT NO. N40085-22-D-0013
TASK ORDER NO. N4008523F6883

for



PUBLIC WORKS DIVISION
MARINE CORPS BASE CAMP
CAMP LEJEUNE, NORTH CAROLINA

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Prepared by:



151 Poole Road; Suite 100
Leland, NC 28451
(910) 383-1044
N.C. License #C-1621

Table of Contents

Introduction	3
Civil scope	3
Structural scope	4
Architectural scope	5
Plumbing scope	6
DOMESTIC WATER CALCULATIONS.....	6
SANITARY SEWER CALCULATIONS	6
HOT WATER CALCULATIONS.....	6
Mechanical scope.....	8
Electrical scope	10
Fire Protection	11
Telecommunications scope	25
Appendix A – Flow Test	
Appendix B – Mechanical Design Analysis	
Appendix C – Electrical Design Analysis	

MRSG DST Storage Facility – MARSOC

Introduction

The project location is an existing developed area at MARSOC. The project consists of a 6,000 SF pre-engineered metal building with standing seam metal roof, brick veneer on steel stud backup, monolithic concrete foundation and floor slab, site work and utilities. Room requirements include a general storage area consisting of 3 cage storage areas with storage racks, overhead door access, and work benches (by others), and toilet room.

Civil scope

DESIGN CRITERIA

The following criteria references shall be used in the design of this project where they apply to the required scope of work:

1. Project RFP
2. UFC 3-201-01 - Civil Engineering
3. AASHTO Assoc. of State Highway & Transportation Officials
4. NCDOT North Carolina Department of Transportation
5. NCDEQ Sedimentation and Erosion Control Manual

Existing Site Conditions

The project site is located north of Anchor Road between Rifle Range Rd on the west and Globe Rd on the east. The site fronts on an existing large, paved parking and access area with existing catch basins. The MRSG building is located directly south of three existing warehouse buildings of similar size and is situated inside of a fenced compound. The site is grassed and the surrounding area to the east is mostly open with grass and tree cover, and several storm ponds.

Demolition will include clearing and grubbing the turfed area, and chain-link fence removal and relocation around the MRSG building to include it within the fenced compound.

Proposed Site and Utility Features

A heavy-duty concrete apron will be provided for vehicular access from the existing paved parking and access area to the building. The concrete apron will consist of a rigid pavement section of 6" of 650 psi flexural concrete with WWF mesh and 6" NCDOT aggregate base course over an approved subgrade. Pavement and sidewalk patches will match existing sections and be performed in a neat manner.

To assist drainage at the impervious areas, a trench drain will be provided along the full width of the loading apron, daylighting to graded swale south of the new building. The building site will be graded to fall towards the naturally sloping undeveloped areas east of the building.

Protective bollards will be utilized at building service doors where vehicular damage could occur.

An existing water distribution line runs in the east-west direction to the south of the site and a new water service and fire service line will be installed to serve the new building. An existing sanitary

sewer collection line runs to the north of the building site in the east-west direction and a new sanitary sewer service line will connect to this main and terminate with a clean-out approximately 5-feet outside the building. These service lines are tapping existing utility mains and NCDEQ water and sanitary sewer permits are not required.

An existing overhead electric line passing over the building site will be relocated by providing a new power pole southeast of the building site. A pad mounted transformer will be provided on site with underground primary provided from the overhead service at the relocated power pole, with underground secondary from the transformer to the building. An existing communication manhole located south of the site will provide communication lines extended to an existing comm handhole near the northeast corner of the building.

All disturbed areas will be sodded.

Stormwater and Erosion Control

The existing drainage patterns on site will be maintained to the maximum extent possible with building downspouts discharging via concrete splash blocks to grade and sheet flowing across the site.

Erosion and sedimentation control measures will be utilized during construction to prevent sediment from leaving the site.

Disturbance on site is limited to less than the required threshold, therefore no environmental permits are expected to be required for this project.

Structural scope

DESIGN CRITERIA

1. Project RFP
2. 2018 International Building Code
3. ASCE 7-16
4. UFC 3-301-01
5. ACI 318-14
6. AISC

Design Loads

All footings and walls will be designed to meet all provided loading included in the RFP specifications/pricing documents provided to the design team.

Anti-Terrorism / Security

No ATRP requirements in structural scope for this project.

Concrete Exposure Class

Concrete exposure class will be F1 (Concrete exposed to freezing-and-thawing cycles with limited exposure to water) as listed in table 19.3.1.1 in ACI 318-14. F1 Exposure Class has a minimum

compressive strength requirement of 3,500 psi and a maximum water/cement ratio of 0.55 per table 19.3.2.1.

Architectural scope

DESIGN CRITERIA

The following criteria references shall be used in the design of this project where they apply to the required scope of work:

1. Project RFP
2. UFC 1-200-01 DoD Building Code, with Change 1
3. UFC 1-200-02 High Performance and Sustainable Building Requirements, with Change 2
4. FC 1-300-09N Navy and Marine Corps Design Procedures, with Change 6
5. UFC 3-101-01 Architecture, with Change 1
6. UFC 3-110-03 Roofing, with Change 5
7. UFC 3-120-01 Design: Sign Standards, with Change 3
8. UFC 3-120-10 Interior Design, with Change 2
9. Architectural Barriers Act Accessibility Standards (ABAAS)
10. 2010 ADA Standards for Accessible Design
11. Marine Corps Base Camp Lejeune Base Exterior Architectural Plan (BEAP)

The architectural scope consists of the design of a new 6,000 sf pre-engineered metal building. The building will include three large open caged areas with storage racks, work benches and a toilet room with shower. There will be 8' wide x 8' high sliding chain link gates between the caged bays.

The exterior of the building will be composed of split face CMU and brick veneer with a steel stud back up. The walls will have 2 ½" rigid insulation or closed cell spray foam insulation in the wall cavity to act as the continuous insulation and moisture barrier for the building. The walls will have a minimum thermal resistance of R-13 insulation in accordance with ASHRAE 90.1. The building will have a reinforced concrete slab on grade with turn down edges and rigid insulation extending 24" horizontally from the building's perimeter. The slab finish will be sealed concrete.

The exterior doors will be insulated electrically operated 10' wide x 12' high metal coiling doors, insulated metal personnel doors and frames. The roof will consist of a hydrostatic prefinished metal roofing system provided by the Pre-Engineered Metal Building Manufacturer. The roof will have an R-19 with an R-11 liner system in accordance with ASHRAE 90.1. There will be a metal panel soffit under the roof overhang.

Aluminum framed translucent panel windows will be placed high in the front and rear walls of the building to allow natural light to the caged areas.

The interior of the building will consist of a large open workspace. There will not be ceilings in the workspace and the floor will be sealed concrete floor with rubber base. This area will house storage lockers, and work bench areas along with other support functions. The toilet and shower room will receive tile floors, full height tile walls, and a painted gypsum board ceiling. There will be

a steel stud framed ceiling platform above the toilet/shower room with a plywood deck. The interior walls will be painted hi-abuse gypsum board on metal stud framing including the perimeter exterior walls. All the storage racks shown on the drawings are for space planning purposes and sizes and modules should be confirmed with the user.

Plumbing scope

DESIGN CRITERIA

UFC 3-420-01 Plumbing Systems (UFC)

PLUMBING SYSTEMS

The plumbing systems within the facilities will consist of the following plumbing fixtures:

- Water Closet
- Lavatory
- Service Sink
- Bi-level Water Cooler
- Wall Hydrant
- Floor Drains
- Water Meter

A reduced pressure zone backflow preventer will be provided on the domestic water service and will be located in the corner of Bay 1- Room 101. Sanitary sewer piping and vent piping for the building will be connected to the site sanitary system at a point approximately 5'-0" outside of the building.

DOMESTIC WATER CALCULATIONS

14 Water Supply Fixture Units (WSFU)
30 GPM, 1-1/4" Dia. Main

SANITARY SEWER CALCULATIONS

11 Drainage Fixture Units (DFSU)
This equates to a 4" sanitary sewer main @ 1/8" per foot slope

HOT WATER CALCULATIONS

Mixed Water Temperature Equation: $P = (T_m - T_c) / (T_h - T_c)$

T_h = Supply hot water temperature

T_c = Inlet cold water temperature

T_m = Desired mixed water temperature

P is the hot water multiplier, which is expressed as a percentage.

$$P = (122 - 60) / (140 - 60)$$

$$P = 62 / 80 = .78$$

$1.75 \text{ GPM (shower demand per head)} \times .78 = 1.37 \text{ GPM}$

$1 \text{ shower} \times 1.37 \times 60 \text{ minutes} = 82.2 \text{ GPH}$

Additional fixtures:

$(1) \text{ Lavatory} \times 6 \text{ GPH} = 6 \text{ GPH}$

$\text{Subtotal} - 6 \text{ GPH} \times .3 \text{ (Demand Factor)} = 1.8 \text{ GPH}$

$\text{Total} - 82.2 + 1.8 = 84 \text{ GPH Peak Demand (1 hour)}$

The water heater specified is 80 gallon (64 gallons available total) with 4.5 kW input and recover a total of 21 GPH at 90-degree rise. Therefore, 85 gallons is available to meet the one-hour peak demand.

FIXTURES AND EQUIPMENT

Water closet will be floor mounted, bottom outlet, dual flush valve, elongated bowl with open front seat less cover. Water usage will be 1.6 / 1.1 gallons per flush.

Lavatory will be wall-hung, vitreous china, with single-handle faucet, 1-1/2 inch P-Trap and 3/8 inch angle supplies with stop valves for hot and cold water. Faucets will be 0.5 gpm flow.

Floor drains will be provided in bathroom and for backflow preventer and water heater.

MATERIALS

The above grade sanitary waste, vent, and storm systems will be PVC pipe and fittings with glued joints. The below grade sanitary waste, vent, and storm systems will be PVC pipe and fittings with glued joints.

Domestic water piping will be copper tubing, ASTM B88; Type L with solder joint fittings fabricated with code approved "lead free" solder for above ground piping and type K copper tubing for below grade applications.

Elastomeric foam insulation with vapor barrier will be applied to all domestic cold and hot water piping and all above grade storm and condensate piping.

Valves will be solder type, ball, for use with the copper tubing, Class 125 in accordance with MSS SP-80.

TESTING

Domestic water piping will be tested in sections at 100 psig for two hours.

DWV piping will be subjected to a final air pressure test of 1.0 inches of water column, or a smoke test. Disinfection will be in accordance with AWWA C651 using a solution containing a minimum of 50 parts per million of available chlorine.

Mechanical scope

DESIGN CRITERIA

The design shall be in accordance with the latest editions of, but not limited to, the following design criteria:

International Mechanical Code – 2015 Edition

UFC 3-401-01 - Mechanical Engineering

UFC 3-410-01 - Heating, Ventilation and Air Conditioning Systems

UFC 4-010-01 - DoD Minimum Antiterrorism Standards for Buildings

ASHRAE 62.1 – 2016 - Ventilation for Acceptable Indoor Air Quality

ASHRAE 90.1 – 2013 - Energy Standard for Buildings except Low Rise Residential Buildings

FC 1-300-09N - Navy and Marine Corps Design Procedures

PROJECT DESCRIPTION

This project consists of one new single story 6,000 sq.ft. facility.

DESIGN CONDITIONS

Outdoor Summer Design temperatures:	91.0°F DB / 77.0°F WB (1% ASHRAE)
Outdoor Winter Design temperature:	27.0°F DB (99% ASHRAE)
Indoor Design Conditions:	Occupied: 76°F Summer / 68°F Winter Unoccupied: 82°F Summer / 55°F Winter
Indoor Design Relative Humidity:	30% to 60%
ASHRAE Climate Zone	3A

MECHANICAL SYSTEM

Each bay storage area will be served by its own split system heat pump. The air handling unit will have MERV 8 and 13 filters on the return/outside air. The amount of outside air will be measured to ensure sufficient quantities are brought into the building to meet ASHRAE 62.1 – 2016 requirements to maintain indoor air quality, and positively pressurize all rooms with exterior openings.

General

Air distribution will be via overhead galvanized sheet metal ductwork designed in accordance with SMACNA Duct Construction Standards. Duct will be wrapped with 2", 1-lb. exterior insulation, no duct liner shall be used. Exposed duct shall be double wall 2" insulated spiral lock seam construction. Exterior duct shall be prefabricated and insulated phenolic duct suitable for exterior installations. The ductwork systems are designed to provide a low noise environment. Air distribution shall be duct or ceiling mounted. Duct construction shall be for low pressure VAV usage.

All air systems shall have a full Test and Balance performed to verify performance match's design.

Exhaust shall be provided in the Toilet room and shall be exhaust directly out the building. Make-up air shall be through a supply grille in the ceiling grid as well as a transfer grille from the storage room.

ATFP

All applicable requirements of UFC 04-010-01 will be followed. All outside air intakes shall be a minimum of 10' above grade. Low leakage dampers will be installed at all building exterior wall penetrations by the HVAC system. An emergency stop button will be located inside the facility to shut down all air moving systems.

BUILDING AUTOMATION SYSTEM

All control and monitoring points will be connected through a single complete building automation system (BAS) suitable for direct digital control (DDC) of all HVAC system components within the building. The control system within the building shall utilize an open protocol implementation of BACnet technology and will utilize ANSI/CEA 709.1B as the communication protocol between DDC hardware devices. All utility meters shall employ the BACnet protocol, with the water meter outputting to the building DDC system and the electric meters outputting directly to the basewide electric metering system. The DDC system will have the ability to limit the operating hours of any equipment, as well as perform setback functions when parts of the building are unoccupied. The DDC system will be compliant with all Cybersecurity requirements. A local display panel for monitoring and scheduling program changes will be provided. The BAS shall be provided with a router to communicate all building points to the basewide EMCS.

CALCULATIONS

- Included in the separate document “MRSG – Mechanical Design Analysis.pdf” are the following:
 - Carrier HAP Load Calculations, including all input data, equipment sizing calculations and outside air calculations.

SUSTAINABLE DESIGN

The HVAC systems for this project have been designed in accordance with UFC 1-200-02. An Energy Model has been used to show that the facility has achieved energy savings above a baseline system. The Energy Model has been run utilizing the Carrier HAP program, version 5.11. The selected system was compared with the ASHRAE 90.1-2010 Appendix G Baseline system, with savings calculated below.

Electrical scope

DESIGN CRITERIA

UFC 3-501-01 Electrical Engineering
UFC 3-520-01 Interior Electrical Systems
UFC 3-530-01 Interior and Exterior Lighting Systems and Controls
UFC 3-550-01 Exterior Electrical Power Distribution
UFC 3-560-01 Electrical Safety, O&M
UFC 3-575-01 Lightning and Static Electricity Protection Systems
NFPA 70 National Electric Code
NFPA 101 Life Safety Code
NFPA 780 Standard for the Installation of Lightning Protection Systems
Applicable UL standards
Applicable NEMA standards

EXTERIOR ELECTRICAL SYSTEMS

A new electrical system will connect the existing 12.47 kV primary on site and (3) new 37.5 kVA pole-mounted transformers will be provided to serve this building. Secondary service conductors shall be fed from these transformers to a service-entrance rated, 208 volt, 3-phase, 100-amp main breaker panel inside the building. A spare conduit will be installed in the service duct bank to allow for future flexibility.

INTERIOR ELECTRICAL SYSTEMS

The electrical building distribution system shall be designed for the type of loads served. All panels shall be fully rated for fault current and designed with a minimum of 20% spare breaker capacity and 15% space in each panel. All panels shall have copper bus, a separate grounding bus, and will utilize bolt-on type breakers. All feeders shall be sized for a maximum voltage drop of 2%, while branch circuits shall not exceed a voltage drop of 3%. Panels that supply non-linear loads will be provided with a 200% rated neutral. All conductors shall be copper, and MC cabling is not allowed. All conductors shall be installed in conduit with a minimum conduit size of 3/4". All electrical devices in this facility shall be U.L. listed and labeled.

Interior lighting shall be designed in accordance with the Illuminating Engineering Society (IES) recommendations. Lighting levels shall meet UFC and user requirements. All new light fixtures shall be energy-efficient LED type. Lighting shall consist of fixtures suitable for the environment in which they are being installed. Emergency lighting shall be separate wall pack fixtures per base standards and shall be spaced to comply with NFPA 101. Test switches shall be integral to fixture. Exit signs shall be LED type, and night lights shall be designed and located according to code requirements. Interior lighting shall be provided at 3500K and exterior lighting at 4000K. Lighting controls, where applicable, will be utilized to meet ASHRAE and UFC requirements, and also to

maximize energy savings. This could include dual technology occupancy/vacancy sensors, relay panel, timeclock, and/or photocell.

Receptacles and power for all areas shall follow project and user requirements as it relates to locations, spacing, heights, and type. Connections for all equipment shall be provided for all new equipment per information provided. Dedicated circuits will be run as required for equipment or user needs. Electrical power and infrastructure for mechanical, plumbing, telecommunications, audio/visual (AV), and other low voltage systems will be coordinated and provided to meet project specifications.

Fire Protection

DESIGN CRITERIA

The following building and fire safety codes and standards apply to this project:

- Unified Facilities Criteria, including, but not limited to:
- UFC 1-200-01, DoD Building Code, 12 June 2023;
- UFC 3-520-01, Interior Electrical Systems, 12 April 2021;
- UFC 3-600-01, Fire Protection Engineering for Facilities, 6 May 2021;
- UFC 4-440-01, Warehouses and Storage Facilities, 1 April 2019
- ICC International Building Code – 2021 Edition (IBC);
- National Fire Protection Association (NFPA) Standards, including, but not limited to:
- NFPA 1, Fire Code – 2024 Edition;
- NFPA 13, Installation of Sprinkler Systems – 2022 Edition;
- NFPA 24, Installation of Private Fire Service Mains and Their Appurtenances – 2022 Edition;
- NFPA 70, National Electrical Code – 2023 Edition;
- NFPA 72, National Fire Alarm and Signaling Code – 2022 Edition;
- NFPA 90A, Installation of Air-Conditioning and Ventilating Systems, – 2024 Edition;
- NFPA 101, Life Safety Code – 2024 Edition (LSC); and
- The Architectural Barriers Act (ABA).

UFC 3-600-01 generally requires construction issues to comply with IBC criteria and life safety/egress issues to meet the requirements of LSC. Fire resistance ratings are required to comply with IBC criteria. For occupancy separations and LSC criteria for interior non-bearing walls (other than occupancy separations) [UFC 3-600-01 §3-2 & 3-3].

DESIGN ANALYSIS

Building Description

The MRSB DST Storage Facility consists of a 6,000-sf one-story building. The primary use of the facility includes storage and support spaces (e.g., toilets). The building will be a pre-engineered metal building. The facility will be equipped with automatic sprinkler protection throughout the building. Code allowances for fully-sprinklered buildings will be applied.

Occupancy

1. The primary use of the building will be for storage.
2. For IBC compliance the facility will be classified as a Group S-1, Moderate Hazard Storage [IBC §311.2] occupancy.
3. For LSC compliance, the facility will be classified as a storage [LSC Chapter 42] occupancy [LSC §6.1.13.1 & LSC §42.1.5].
4. Special hazard areas including any mechanical, utility, and electrical areas are considered incidental to the main occupancy. These areas are required to be protected in accordance with LSC Section 8.7.1.

Construction Requirements

1. The minimum construction type of a structure is required to comply with the fire resistance requirements and fire area limitations in UFC 3-600-01 and the IBC. The minimum construction type of a building is typically based on the height, area, occupancy, open street frontage, and automatic sprinkler protection provided in the building.
2. In accordance with IBC Chapter 5, the building will be constructed to be Type IIB construction. The height and area limitations in Table 3.1 apply to the building for confirming the minimum required construction type.

Table 3.1: Limitations for Type IIB Construction

	Group S-1 Single-Story Sprinklered
Allowable Height [IBC Table 504.3]	75-ft

Allowable Number of Stories [IBC Table 504.4]*	1
Allowable Area Per Story [IBC Table 506.2]	70,000-sf

*: Limited to a single story to allow for the increase in allowable area for a single-story building per IBC Table 506.2.

3. The building has a total footprint area of approximately 6,000-sf. As the allowable area for Group S-1 of 70,000-sf is greater than the actual footprint area of the building Type IIB construction is allowable.

4. The required fire resistance ratings of building structural elements as specified in IBC Table 601 are provided in Table 3.2 below for Type IIB construction:

MRSO DST Storage Facility – MARSOC November 30, 2023

Camp Lejeune, North Carolina p. 5

Fire Protection Design Analysis 50% Submittal

Table 3.2: Type IIB Resistance Ratings of Structural Elements

Building Element	IBC Type IIB
Structural frame, including columns, girders, trusses ^a	0-hr
Bearing walls, exterior ^b	0-hr
Bearing walls, interior ^c	0-hr
Non-bearing walls, exterior ^b	0-hr
Non-bearing walls, interior ^c	0-hr
Floor construction, including supporting beams and joists	0-hr
Roof construction, including supporting beams and joists	0-hr

^a The structural frame is considered to be the columns and the girders, beams, trusses, and spandrels having direct connections to the columns and bracing members designed to carry gravity loads.

^b Based on IBC Table 705.5 for a fire separation distance of 10-feet or more.

^c Not less than the fire resistance required by other section of the code.

5. The roof coverings and roof deck assemblies must be in accordance with UFC 3-600-01 Sections 15-1 and 15-2.

Exterior Fire Exposure

1. The requirements for providing fire resistance-rated exterior walls and openings are based on the fire separation distances (FSD) between buildings and the occupancy groups within the buildings.

2. The fire separation distance is defined as the distance measured from the building face to the

closest interior lot line, to the centerline of a street or public way, or to an imaginary line between two buildings on the same lot [IBC §602].

3. The fire separation distance is a measure of the separation to a point along a line between buildings, not the actual distance between buildings. The fire separation distance is measured at right angles from the face of the wall.

4. Once the location of the lot line is established, all future provisions must be based on this lot line and applied to both buildings. Where the lot line is not parallel to the exterior wall of a building, the required fire resistance rating may vary along the building side.

5. For building sides not fronted by a street, the fire separation distance is measured to an imaginary line between the buildings assuming the buildings are located on the same property (i.e., an interior lot line).

6. Where a new building is erected on the same lot as an existing building, the code requires the assumed property line to be located such that the existing building complies with exterior wall and opening protection criteria [IBC §605.3].

7. The fire-resistance rating of exterior walls with a fire separation distance of 10-ft or less shall be rated for exposure to fire from both sides [IBC §705.5].

8. The exterior walls are required to extend the full height of the building and be constructed to remain in place for the duration of time indicated by the required fire-resistance rating [IBC §705.6].

9. The code requirements for the fire-resistance rating and exterior wall openings are provided in Table 3.3.

Table 3.3: Fire-Resistance Rating Requirements for Exterior Walls of Type IIB Buildings Based on Fire Separation Distance [IBC Table 705.5]

Fire Separation Distance (FSD)	Group S-1
FSD < 5-feet	2-hr
5-feet ≤ FSD < 10-feet	1-hr
FSD ≥ 10-feet	r

10. The Storage Facility is at least 40-ft from the adjacent buildings and any other structure on all

sides. This provides a FSD of at least 30-ft from the adjacent buildings to an imaginary lot line and at least 10-ft from the imaginary lot line to the Storage Facility. The exterior wall of the building is not required to have a fire-resistance rating and there is no limit on the area of openings [IBC §705.8.1 Exception 2].

11. Openings in the exterior wall are required to be in accordance with IBC Section 705.8.3.

HVAC Systems

1. Air handling, HVAC, and exhaust systems, including fire/smoke dampers, are required to comply with NFPA 90A except as modified by UFC [UFC 3-600-01 §28-1].

2. Fire dampers are required in ductwork or air transfer openings that penetrate fire resistance rated assemblies. NFPA 90A permits the deletion of fire dampers in fully ducted penetrations of 1-hour fire resistance-rated partitions.

3. The installation of duct detectors is required to comply with NFPA 90A. Smoke detectors listed for use in air distribution systems shall be located downstream of the air filters and ahead of any branch connections in air supply systems having a capacity greater than 2,000-cfm [NFPA 90A §11.4.2.1 (1)].

4. Plenum spaces may be used as part of an air handling system where the plenums comply with NFPA 90A [UFC 3-600-01 §28-1.2]. Rooms or areas that form a plenum space or that are used as a plenum must not be occupied for any purpose except during repairs and maintenance operations to the air-handling equipment [UFC 3-600-01 §28-1.2].

Interior Finish Materials

1. Interior walls and ceiling finishes are required to comply with UFC 3-600-01 Section 8-1 which requires wall and ceiling finishes, and movable partitions to conform to the requirements of the LSC. The LSC restricts the use of certain materials as interior finishes on walls and ceilings based on the flame spread rating and smoke development rating of the material. Interior wall and ceiling finishes shall be classified in accordance with ASTM E-84.

2. The interior finish materials shall be grouped in the following classes, based on their flame

spread rating and smoke developed rating [LSC §10.2.3.4]:

Class A – Flame spread 0-25, smoke developed 0-450

Class B – Flame spread 26-75, smoke developed 0-450

Class C – Flame spread 76-200, smoke developed 0-450

3. Table 3.5 summarizes the interior finish requirements for the Storage Facility [LSC Table A.10.2.2 note (6)].

Table 3.5: Minimum Interior Finish Requirements

Occupancy	Exits	Exit Access Corridors	Other Spaces
Storage	Minimum Class C	Minimum Class C	Minimum Class C

Note: Automatic sprinkler protection allowances have been applied.

4. The codes also specify requirements for interior floor finish and floor covering materials. Floor coverings other than carpet shall have a minimum critical radiant flux of 0.1 W/cm² [LSC §10.2.7.2]. Interior floor finish materials shall be classified in accordance with NFPA 253, Standard Method of Test for Critical Radiant Flux for Floor Covering Systems Using a Radiant Heat Energy Source as follows [LSC §10.2.7.4]:

Class I – Critical radiant heat flux ≥ 0.45 W/cm²

Class II – Critical radiant heat flux ≥ 0.22 W/cm²

5. There are no additional requirements for floor finish in all areas for storage occupancies [LSC Table A-10.2.2, Note 6].

Means of Egress

Means of egress are required to comply with the LSC except as modified by the UFC [UFC 3-600-01 §10-

1.1]. Where egress elements serve multiple occupancies, the most restrictive requirements apply.

Occupant Load

1. The calculated occupant loads are required to be determined in accordance with LSC Section 7.3.1.2. The occupant load is determined based on the use and area of the space, and the

occupant load factors specified by the LSC. Table 3.6 summarizes the applicable occupant load factors for the Storage Facility.

Table 3.6: Applicable Occupant Load Factors

Use Occupant Load Factor (sf/person)
Storage Maximum Probable Business 150

2. Table 3.7 summaries the occupant load calculations for the building per the LSC.

Table 3.7: Calculated Occupant Loads per LSC

Area	Occupancy / Use	Approx. Area (sf)	Occupant Load Factor (sf/person)	Occupant Load (persons)
Toilet	Business	93	150	1
Bay 1	Storage	1,708	Max. Probable	4
Bay 2	Storage	1,887	Max. Probable	4
Bay 3	Storage	1,825	Max. Probable	4
Total		5,513	-	13

Number & Capacity of Means of Egress

1. A minimum of two means of egress is required from areas with an occupant load of 500 persons or less, a minimum of three means of egress is required for areas with an occupant load of 500 to 1,000 persons, and a minimum of four means of egress is required for areas with an occupant load of more than 1,000 persons [LSC §7.4.1.2, §42.2.4.1].
2. A single means of egress is permitted when the travel distance is less than the common path of travel limitation [LSC §42.2.4.1(2)].
3. The LSC specifies an exit capacity factor 0.2 inches of exit width per person for level exit components (e.g., doors, corridors, ramps) [LSC Table 7.3.3.1].
4. Table 3.8 summarizes the number and capacity of exits provided for the renovation areas per the LSC.

Table 3.8: Number of Exits and Capacity

Floor Level	Required Exit Capacity	Available Exit Capacity	Number of Exits Required	Number of Exits Provided
Storage Facility	13	680	1	4

5. Based on the proposed floor plan, the number and capacity of the exits serving the Storage Facility are adequate.

Arrangement of Means of Egress

1. Where two or more means of egress (exits or exit access) are required, the means of egress are required to be located remote from one another, so that the distance between them is at least one-half the maximum diagonal distance of the area served [LSC §7.5.1.3.2].
2. Travel distance to an exit, dead-ends, and common paths of travel are limited based on occupancy [LSC Table A.7.6]. Table 3.9 summarizes the exit access travel distance, dead-end, and common path of travel limitations for the facility.

Table 3.9: General Arrangement of Means of Egress Criteria

Occupancy	Exit Access Travel Distance (ft)	Common Path of Travel Limit (ft)	Dead-end Corridor Limit (ft)
Storage	400	100	100

Note: (1) Automatic sprinkler protection allowances have been applied.

3. Travel distance, common path of travel, and dead ends will be provided in accordance with LSC Table A.7.6.

Accessible Means of Egress

1. Accessible means of egress shall comply with LSC [UFC 3-600-01 §10-1.2]. Areas accessible to people with severe mobility impairment are required to be provided with not less than two accessible means of egress unless exit access travel along the accessible means of egress is less than the permitted common path of travel [LSC §7.5.4.1 & §7.5.4.1.4].
2. Where two accessible means of egress are required, the exits serving such means of egress will be located at a distance from one another not less than one-half the length of the maximum overall diagonal of the building/area unless the building is fully sprinklered [LSC §7.5.4.2 & §7.5.4.2.2].
3. Accessible means of egress will be continuous from each accessible occupied area to a public way or area of refuge [LSC §7.5.4.3].

4. The Storage Facility is classified as a one-story Storage Occupancy; therefore, the accessible means of egress will consist of the following component:

a. First Floor

- i. Fully sprinklered floor
- ii. Access to the public way

Exit Discharge

1. All exits will discharge directly to the outside [LSC §7.7.1].

Egress Doors

1. Doors serving as part of the means of egress are required to comply with LSC Section 7.2.1. Door openings in means of egress are required to be not less than 32-in clear width [LSC §7.2.1.2.3].

Where a pair of doors is provided, not less than one of the doors is required to provide at least a 32-in clear width.

2. Doors in the means of egress are generally required to be of the side-hinged or pivoted-swinging type [LSC §7.2.1.4].

3. Where serving a room or area with an occupant load of 50 or more, doors are required to swing in the direction of egress travel [LSC §7.2.1.4.2].

4. Doors shall be arranged to be opened readily from the egress side whenever the building is occupied. Locks, if provided, shall not require the use of a key, a tool, or special knowledge or effort for operation from the egress side [LSC §7.2.1.5].

5. Access-controlled egress doors (e.g., equipped with card readers) must comply with LSC Section 7.2.1.6.2. The LSC requirements do not practically permit egress in two directions through doors equipped with access-control hardware.

Automatic Suppression Systems

1. Section 4-48.2.1 of UFC-3-600-01 requires automatic sprinkler protection in warehouses and storage facilities, regardless of floor area or construction type. As such, sprinkler protection is

required throughout the facility. The sprinkler system must meet the requirements of UFC 3-600-01, UFC 4-440-01, and NFPA 13.

2. All sprinkler system piping must be installed in compliance with UFC 3-600-01 Section 9-7.6. Pipe 2-in and less in diameter must be minimum Schedule 40. Pipe larger than 2-in in diameter must be minimum Schedule 10 [UFC 3-600-01 §9-7.6.2]

3. The incoming sprinkler service will be provided with a reduced pressure backflow prevention assembly.

4. Quick response sprinklers will be provided throughout. Upright sprinklers will be provided in rooms without finished ceilings. Recessed sprinklers will be provided in rooms with finished ceilings. All sprinklers will have ordinary temperature ratings unless more stringent requirements apply.

5. Fire department connections shall follow the requirements in NFPA 13 Section 16.12.3. Fire department connections shall consist of one 5-in Storz connection. Fire department connections for sprinkler systems will be provided within 150-ft of fire hydrants.

6. Per UFC 3-600-01 Section 4-48.2.3 sprinkler systems for warehouses shall be provided per NFPA 13. Design requirements for these configurations are contained in Table 3.10. The remaining areas of the building will be classified as either Light or Ordinary Hazard.

Table 3.10: Sprinkler System Storage Design Criteria

Room Description	NFPA 13 Code Reference	Storage Description	Design Criteria
Warehouse (ST)	Table 21.4.1.2.1.1(a)	Class IV Commodities High Temp. Sprinklers Single- & Double-Row Racks up to 14'	0.30-gpm/sf over 2,000-sf 500-gpm hose stream 120-min

7. Preliminary hydraulic calculations are based on the most hydraulic demanding area – single- & double-row rack storage of Class IV Commodities with a storage height of 14-ft.

Water Supply

The fire protection water supply, including fire hydrants, serving the buildings is required to comply with

UFC 3-600-01 Section 9-6.4. The water supply data from flow tests performed on August 23, 2023, is

shown in Table 3.11.

Table 3.11: Summary of Water Supply Tests

	Measurement
Static Pressure	78-psi
Residual Pressure	70-psi
Flow	1,325-gpm

Fire Department (Emergency) Vehicle Access

1. Fire department vehicle access is required to comply with UFC 3-600-01 Section 9-1.1. All buildings greater than 5,000-sf, or more than two stories in height, are required to have at least one means of all-weather ground access to allow emergency vehicles unimpeded access to the building. The all-weather ground access is required to be paved, start from the road, and terminate no farther than 33-ft from a door from the building. Where approved by the AHJ, an engineered all-weather surface that is not paved may be provided.
2. Facilities with fire department connections for sprinkler systems must be provided with suitable all-weather ground access surface for pumper apparatus within 150-ft of such fire department connections.
3. The fire department access shall meet the dimensions for fire lanes and turnarounds contained in NFPA 1. All force protection equipment, such as bollards or gates, must not require more than one person to remove or open. Access may require fire apparatus to drive over a curb. Any locking device controlling vehicle access must be under control of the Fire Department or 24-hr security personnel located at the facility.

Fire Hydrants

1. Fire hydrants must be provided to protect the facility. All parts of the Storage Facility must be within 350 ft of a hydrant with consideration given to accessibility and obstructions [UFC 3-600-01 §9-3.5.8.2]. Hydrants protecting warehouses must be spaced at 300 ft intervals [UFC 3-600-

01 §9-3.5.8.4.b]. Hydrants must be installed adjacent to paved areas, accessible to fire department apparatus [UFC 3-600-01 §9-3.5.7.b].

Fire Alarm System

1. A complete, electrically supervised, addressable intelligent, manual, automatic, annunciated fire alarm system shall be provided throughout the building. This system will be installed in accordance with UFC 3-600-01 and NFPA 72.
2. The Fire Alarm Control Panel (FACP) shall be located in Bay 1.
3. Audible notification is required to be provided within all occupied spaces.
4. Clear strobes marked with the word “ALERT” will be provided. Visible notification is required as follows:
 - a. In normally occupied, public and common use areas (e.g. break rooms, corridors, or conference rooms).
 - b. In all normally unoccupied areas (such as mechanical rooms, electrical rooms, janitor rooms, storage areas, communications closets, and other similar spaces) greater than 900-sf.
 - c. In any normally unoccupied area where the ambient noise is loud enough to require hearing protection.
 - d. In all offices that are designed for, or may contain, more than four persons at any one time.
5. The fire alarm system design and installation will include initiation of the fire alarm system by actuation of any sprinkler system water-flow devices, smoke detectors, carbon monoxide detectors, or manual pull stations.
6. Smoke detection is required in HVAC ducts in accordance with NFPA 90A.
7. Carbon monoxide detection shall be required in all rooms with fuel-burning equipment [UFC 3-600-01 §5-9.1] and in each HVAC zone served by a separate air handler [UFC 3-600-01 §5-9.2].
8. Manual fire alarm pull stations will be provided at each building exit.
9. The fire alarm system is required to interface with the lighting control system to energize all means of egress lighting controlled by occupancy sensors upon activation of the building fire

alarm.

10. The system is required to be connected to a central alarm location, fire department, or alarm monitoring location [UFC 3-600-01 §5-2.1.1]. The fire reporting portion of the system shall report to the central fire alarm reporting station via a Digital Alarm Communicator Transmitter (DACT).

11. Class B notification appliances, signaling line circuits, and initiating device circuits will be provided and installed in conduit.

12. If more than one fire protection circuit is required, a panelboard dedicated to fire protection and life safety systems will be provided [UFC 3-520-01 §3-2.7.a].

Fire Extinguishers

1. Fire extinguishers shall be provided in accordance with LSC [UFC 3-600-01 §9-17.1]. Fire extinguishers are not required in Storage occupancies [NFPA 101 §42.3.5].

Emergency Lighting & Exit Signage

1. All means of egress are required to be illuminated in accordance with LSC Section 7.8 [LSC §42.2.8.1].

a. Exit access will include only designated aisles, corridors, and passageways leading to an exit [LSC §7.8.1.1]. Exit discharge will include only designated aisles, corridors, walkways, and passageways leading to a public way [LSC §7.8.1.1].

b. Artificial lighting is required at locations and times necessary to maintain adequate illumination [LSC §7.8.1.2.1].

c. When automatic controls are used to temporarily turn off the illumination within the means of egress, the system must comply with the criteria provided in LSC Section 7.8.1.2.2. The fire alarm system is required to interface with the lighting control system to energize all means of egress lighting controlled by occupancy sensors upon activation of the building fire alarm.

2. Emergency lighting facilities are required for the means of egress [LSC §42.2.9]. The emergency

lighting system is required to provide continued illumination for a duration of not less than 90 minutes in case of emergency or primary power loss [LSC §7.9.2.1].

3. Exit signage is required for exits and exit access routes in accordance with LSC Section 7.10 and UFC-3-600-01 Section 10-2 [LSC §42.2.10]. Signs must have lettering on an opaque background. Internally illuminated signs must be light emitting diode (LED) type, electroluminescence (LEC), or cold cathode type. Incandescent fixtures are not permitted. Radioluminous exit signs are not permitted [UFC 3-600-01 §10-2.2].

CONCLUSION

The MRSG DST Storage Facility will comply with UFC-3-600-01 and National Fire Codes.

APPENDIX A – HYDRAULIC CALCULATIONS

Hydraulic Analysis Worksheet

MCB CL MRSG DST Storage Facility – Double-Row Rack Storage (Class IV)

1. Water Quantity:

- a. $\text{Density} \times \text{Design Area} \times 1.2 = 720.0 \text{ gpm}$
Includes 20% overage
- b. Outside Hose Stream Allowance = 500.0 gpm
- c. Other = 0.0 gpm
- d. Domestic Demand = 0.0 gpm

Total Water Demand (a+b+c+d) = 1220.0 gpm

2. Water Pressure:

- a. End-head-pressure = 14.1 psi
 $[\text{density} \times (\text{area}/\text{head}) / \text{k-factor}]^2$
- b. Elevation Loss = height $\times 0.433 = 9.5 \text{ psi}$
- c. Outside Friction Loss = 3.0 psi
- d. Backflow Preventer = 12.0 psi
RPZ Backflow Preventer
- e. Inside Friction Loss = 20.0 psi

Total Pressure (a+b+c+d+e) + 10% SF = 64.4 psi

3. Available Pressure:

$P_a = P_s - (P_s - P_r) \times (Q_a / Q_t)^{1.85} = 71.1 \text{ psi}$

Available Pressure Greater than Required Total

Telecommunications scope

DESIGN CRITERIA

UFC 3-580-01 Telecommunications Building System Design
Camp Lejeune Base Specification 27 10 00
Camp Lejeune Base Specification 33 82 00
Applicable EIA and TIA standards
Applicable UL standards
NFPA 70 National Electric Code

EXTERIOR TELECOMMUNICATIONS SYSTEMS

This facility shall be provided with new OSP infrastructure and service to include OSP copper and single mode OSP fiber. The planned cable counts are 25 pair copper and 12 strand SM FOC.

OSP cabling point of connection is existing manhole MHRR110-3-5 and the duct bank point of connection is existing manhole MHRR110-3-6. Both existing manholes are nearby on the other side of C Street. All new duct bank shall be concrete-encased.

INTERIOR TELECOMMUNICATIONS SYSTEMS

The new telecommunications service for each building shall include, but not be limited to, vertical, horizontal, and backbone distribution, copper and fiber optic backbone and distribution cabling, termination equipment, workstation outlets, racks, cable trays, cable management, patch panels, conduits, backboards, and grounding. Cable trays will be used for cable distribution throughout the building, and ladder racks will be utilized at the telecommunications cabinet. Voice and data connectivity will be provided for all workstations, floor boxes, printers, and projectors per project and user requirements. Cabling shall be CAT6 and rated for its application. All telecommunications requirements shall adhere to the applicable design criteria and base standards.

A telecommunications room is not required in this building, and an enclosed telecommunications cabinet will suffice. This facility also does not have a requirement for SIPRnet.

Appendix B – FIRE FLOW TEST

Hydrant Flow Test Report

Location MCB Camp Lejeune - MRSG DST Storage Facility Site Date 23 Aug 2023

Test made by Base Utilities Time 1030

Representative of MCB Camp Lejeune

Witness Aaron Freid, Jensen Hughes

State purpose of test Determine the available water supply for the MRSG DST Storage Facility.

Consumption rate during test 1.325-gpm

If pumps affect test, indicate pumps operating PUMP

Flow hydrants: A₁ A₂ A₃ A₄

Size nozzle 2.5-in

Pitot reading 62.3-psi

Discharge coefficient 0.9 Total gpm

gpm 1,325 1,325

Static B 78 psi Residual B 70 psi

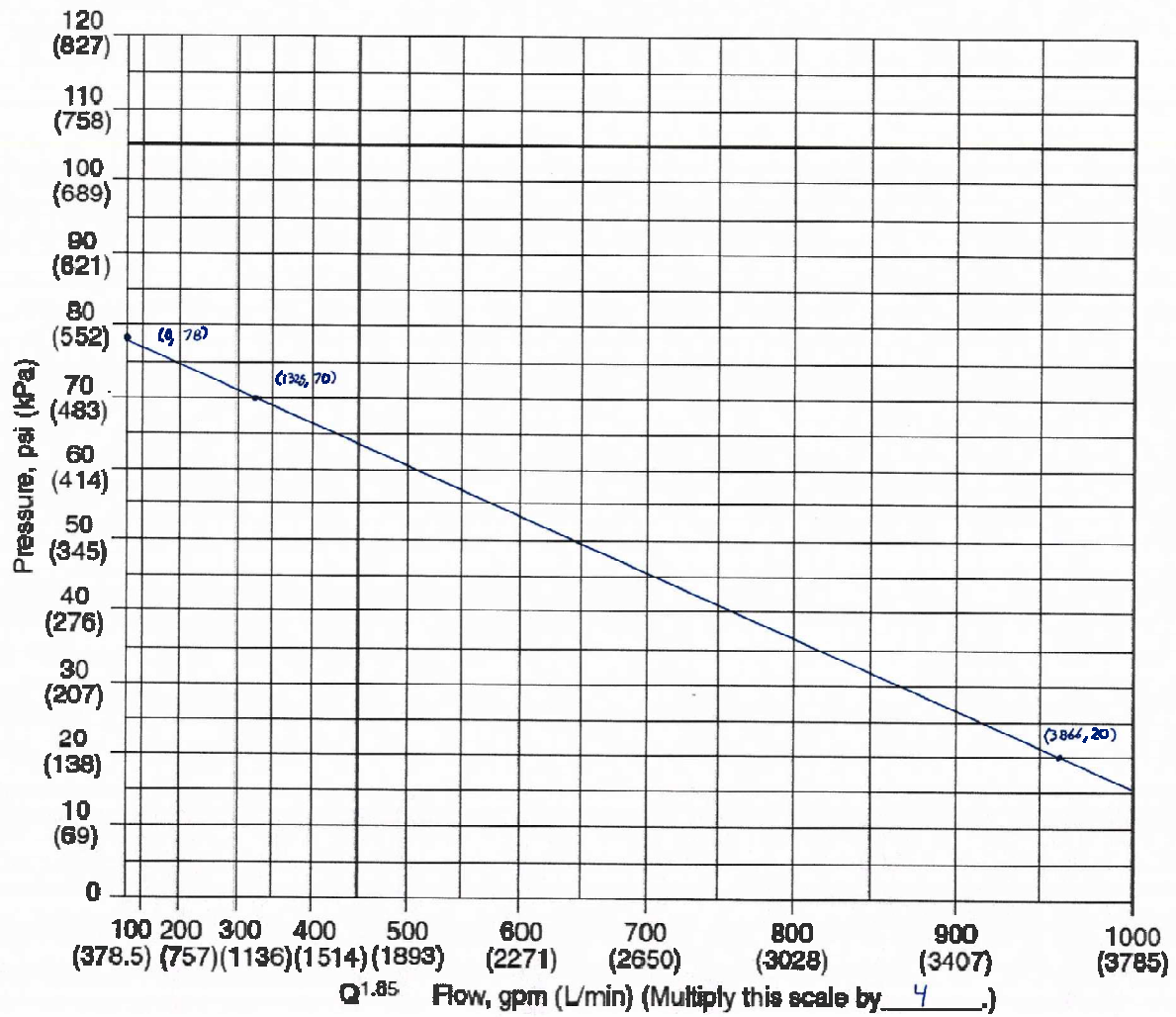
Projected results @20 psi Residual 3,866 gpm; or @ psi Residual gpm

Remarks 12-in main running along W Anchor Rd (formerly C St).

Hydrant A1 (Flow): 10-75-12; Hydrant B (Pressure): 10-76-12

Location map: Show line sizes and distance to next cross-connected line. Show valves and hydrant branch size. Indicate north. Show flowing hydrants – Label A₁, A₂, A₃, A₄. Show location of static and residual – Label B.

Indicate B Hydrant X Sprinkler Other (identify)





Appendix C – MECHANICAL DESIGN ANALYSIS

**CAMP LEJEUNE – MRSG DST STORAGE
FACILITY**

**MECHANICAL DESIGN ANALYSIS
100% SUBMITTAL**

MRSB DST Storage Facility

TABLE OF CONTENTS

- I. Proposed Building HVAC Loads - Inputs**
 - a. Spaces**
 - b. Wall**
 - c. Window**
 - d. Roof**
 - e. Air Systems**
 - f. Building**

- II. Baseline Building HVAC Loads – Inputs**
 - a. Spaces**
 - b. Wall**
 - c. Window**
 - d. Roof**
 - e. Air Systems**
 - f. Building**

- III. Proposed Building HVAC Loads - Outputs**
 - a. Equipment Sizing**
 - b. Outside Air Calculations**

- IV. Baseline Building HVAC Loads - Outputs**
 - a. Equipment Sizing**
 - b. Outside Air Calculations**

- V. Energy Model Inputs**
 - a. Weather Reports**
 - b. Electric Rate Input**
 - c. Fuel Rate Input**

- VI. Proposed vs Baseline Energy Model Outputs**
 - a. Building Simulation Report**

I. Proposed Building HVAC Loads – Inputs

Space Input Data

Camp Lejeune - MRSRG DST Storage
CRENSHAW CONSULTING ENGINEERS, INC.

11-29-2023
04:01PM

Restroom

1. General Details:

Floor Area 82.0 ft²
Avg. Ceiling Height 9.0 ft
Building Weight 70.0 lb/ft²

1.1. OA Ventilation Requirements:

Space Usage User-Defined
OA Requirement 1 0.0 CFM/person
OA Requirement 2 0.00 CFM/ft²
Space Usage Defaults ASHRAE Std 62.1-2010

2. Internals:

2.1. Overhead Lighting:

Fixture Type Recessed (Unvented)
Wattage 60.0 Watts
Ballast Multiplier 1.00
Schedule 90.1 Warehouse Lights/Elec

2.4. People:

Occupancy 0.0 Person
Activity Level Office Work
Sensible 245.0 BTU/hr/person
Latent 205.0 BTU/hr/person
Schedule None

2.2. Task Lighting:

Wattage 0.00 W/ft²
Schedule None

2.5. Miscellaneous Loads:

Sensible 0 BTU/hr
Schedule None
Latent 0 BTU/hr
Schedule None

2.3. Electrical Equipment:

Wattage 0.00 W/ft²
Schedule None

3. Walls, Windows, Doors:

Exp.	Wall Gross Area (ft ²)	Window 1 Qty.	Window 2 Qty.	Door 1 Qty.
W	90.0	0	0	0
S	90.0	0	0	1

3.1. Construction Types for Exposure W

Wall Type MRSG Wall

3.2. Construction Types for Exposure S

Wall Type MRSG Wall
Door Type 2013 Zone 3 Opaque Nonswinging

4. Roofs, Skylights:

(No Roof or Skylight data).

5. Infiltration:

Design Cooling 0.00 CFM
Design Heating 0.00 CFM
Energy Analysis 0.00 CFM
Infiltration occurs only when the fan is off.

6. Floors:

Type Floor Above Conditioned Space
(No additional input required for this floor type).

7. Partitions:

(No partition data).

Space Input Data

Camp Lejeune - MRSRG DST Storage
CRENSHAW CONSULTING ENGINEERS, INC.

11-29-2023
04:01PM

Storage

1. General Details:

Floor Area **5440.0** ft²
Avg. Ceiling Height **10.0** ft
Building Weight **70.0** lb/ft²

1.1. OA Ventilation Requirements:

Space Usage **MISCELLANEOUS: Warehouse**
OA Requirement 1 **10.0** CFM/person
OA Requirement 2 **0.06** CFM/ft²
Space Usage Defaults **ASHRAE Std 62.1-2010**

2. Internals:

2.1. Overhead Lighting:

Fixture Type **Recessed (Unvented)**
Wattage **1152.0** Watts
Ballast Multiplier **1.00**
Schedule **90.1 Warehouse Lights/Elec**

2.4. People:

Occupancy **4.0** People
Activity Level **Medium Work**
Sensible **295.0** BTU/hr/person
Latent **455.0** BTU/hr/person
Schedule **90.1 Warehouse Occupants**

2.2. Task Lighting:

Wattage **0.00** W/ft²
Schedule **None**

2.5. Miscellaneous Loads:

Sensible **0** BTU/hr
Schedule **None**
Latent **0** BTU/hr
Schedule **None**

2.3. Electrical Equipment:

Wattage **0.25** W/ft²
Schedule **90.1 Warehouse Lights/Elec**

3. Walls, Windows, Doors:

Exp.	Wall Gross Area (ft ²)	Window 1 Qty.	Window 2 Qty.	Door 1 Qty.
N	1584.0	0	0	0
S	1584.0	0	0	4
W	726.0	0	0	0
E	907.0	0	0	0

3.1. Construction Types for Exposure N

Wall Type **MRSG Wall**

3.2. Construction Types for Exposure S

Wall Type **MRSG Wall**
Door Type **2013 Zone 3 Opaque Nonswinging**

3.3. Construction Types for Exposure W

Wall Type **MRSG Wall**

3.4. Construction Types for Exposure E

Wall Type **MRSG Wall**

4. Roofs, Skylights:

(No Roof or Skylight data).

5. Infiltration:

Design Cooling **0.00** CFM
Design Heating **0.00** CFM
Energy Analysis **0.00** CFM
Infiltration occurs only when the fan is off.

6. Floors:

Type **Floor Above Conditioned Space**
(No additional input required for this floor type).

7. Partitions:

(No partition data).

Wall Constructions

Camp Lejeune - MRSG DST Storage
CRENSHAW CONSULTING ENGINEERS, INC.

11-29-2023
04:01PM

MRSG Wall

Wall Details

Outside Surface Color **Medium**
Absorptivity **0.675**
Overall U-Value **0.046** BTU/(hr·ft²·°F)

Wall Layers Details (Inside to Outside)

Layers	Thickness in	Density lb/ft ³	Specific Ht. BTU / (lb·°F)	R-Value (hr·ft ² ·°F)/BTU	Weight lb/ft ²
Inside surface resistance	0.000	0.0	0.00	0.68500	0.0
6" STEEL STUDS	6.000	490.0	0.12	1.75000	245.0
3" Rigid Insulation	3.000	2.0	0.22	18.60000	0.5
4-in face brick	4.000	125.0	0.22	0.43290	41.7
Outside surface resistance	0.000	0.0	0.00	0.33300	0.0
Totals	13.000	-		21.80090	287.2

Window Constructions

Camp Lejeune - BEQ HP505
CRENSHAW CONSULTING ENGINEERS, INC.

11-06-2023
04:45PM

HP514 - 4'

Window Details:

Detailed Input	No	
Height	4.00	ft
Width	1.00	ft
Overall U-Value	0.400	BTU/(hr·ft ² ·°F)
Overall Shade Coefficient	0.400	

Roof Constructions

Camp Lejeune - MRSB DST Storage
CRENSHAW CONSULTING ENGINEERS, INC.

11-29-2023
04:02PM

2013 Zone 2-3 Insulation Entirely Above Deck

Roof Details

Outside Surface Color **Medium**
Absorptivity **0.675**
Overall U-Value **0.039** BTU/(hr·ft²·°F)

Roof Layers Details (Inside to Outside)

Layers	Thickness in	Density lb/ft ³	Specific Ht. BTU / (lb·°F)	R-Value (hr·ft ² ·°F)/BTU	Weight lb/ft ²
Inside surface resistance	0.000	0.0	0.00	0.61000	0.0
22 gage steel deck	0.034	489.0	0.12	0.00011	1.4
Board Insulation	2.900	2.0	0.22	25.00000	0.5
Outside surface resistance	0.000	0.0	0.00	0.17000	0.0
Totals	2.934	-		25.78011	1.9

[P] AHU Input Data

Project Name: Camp Lejeune - MRSRG DST Storage
Prepared by: CRENSHAW CONSULTING ENGINEERS, INC.

11-29-2023
04:02PM

1. General Details:

Air System Name [P] AHU
Equipment Type Split AHU
Air System Type Single Zone CAV
Number of zones 1

2. Ventilation System Components:

Ventilation Air Data:

Airflow Control Constant Ventilation Airflow
Ventilation Sizing Method ASHRAE Std 62.1-2010
Unocc. Damper Position Closed
Damper Leak Rate 0 %
Outdoor Air CO2 Level 400 ppm

Central Cooling Data:

Supply Air Temperature 58.0 °F
Coil Bypass Factor 0.100
Cooling Source Air-Cooled DX
Schedule JFMAMJJASOND
Capacity Control Cycled or Staged Capacity - Fan On

Central Heating Data:

Supply Temperature 95.0 °F
Heating Source Air Source Heat Pump
Schedule JFMAMJJASOND
Capacity Control Cycled or Staged Capacity - Fan On

Supply Fan Data:

Fan Type Forward Curved with Variable Frequency Drive
Configuration Draw-thru
Fan Performance 2.50 in wg
Overall Efficiency 50 %
Fan Control 2-speed fan cooling, 1-speed fan heating
Low Speed Fan Airflow 67 %

Duct System Data:

Supply Duct Data:

Duct Heat Gain 0 %
Duct Leakage 0 %

Return Duct or Plenum Data:

Return Air Via Ducted Return

3. Zone Components:

Space Assignments:

Zone 1: Zone 1	
Restroom	x1
Storage	x1

Thermostats and Zone Data:

Zone	Cooling T-Stat Occ. (°F)	Cooling T-Stat Unocc. (°F)	Heating T-Stat Occ. (°F)	Heating T-Stat Unocc. (°F)	T-Stat Throttling Range (°F)	Diversity Factor (%)	Direct Exhaust Airflow (CFM)	Direct Exhaust Fan (kW)
1	75.0	80.0	70.0	65.0	1.50	100	0.0	0.0

Thermostat Schedule 90.1 Warehouse HVAC
Unoccupied Cooling is Available

Supply Terminals Data:

Zone	Terminal Type	Air Distribution	Air Distribution Effectiveness Specification	Air Distribution Effectiveness	Minimum Airflow	Fan Performance	Fan Efficiency	Design Supply Temp.
1	Diffuser (no reheat)	Ceiling supply / ceiling return	ASHRAE 62.1	<auto>	0.00 CFM/person	-	-	-

[P] AHU Input Data

Project Name: Camp Lejeune - MRSB DST Storage
Prepared by: CRENSHAW CONSULTING ENGINEERS, INC.

11-29-2023
04:02PM

Zone Heating Units:

Zone Unit Heat Source **Electric Resistance**
Zone Heating Unit Schedule **JFMAMJJASOND**

4. Sizing Data (Computer-Generated):

System Sizing Data:

Sizing Data:

Cooling Supply Temperature **58.0** °F
Supply Fan Airflow **574.8** CFM
Ventilation Airflow **461.3** CFM
Heating Supply Temperature **95.0** °F

Hydronic Sizing Specifications:

Chilled Water Delta-T **10.0** °F
Hot Water Delta-T **20.0** °F

Safety Factors:

Cooling Sensible **0** %
Cooling Latent **0** %
Heating **0** %

Zone Sizing Data:

Zone Airflow Sizing Method **Sum of space airflow rates**
Space Airflow Sizing Method **Individual peak space loads**

Zone	Supply Airflow (CFM)	Zone Htg Unit (MBH)	Reheat Coil (MBH)	- (CFM)
1	574.8	-	-	

5. Equipment Data

Central Cooling Unit - Air-Cooled DX

Estimated Maximum Load **39.2** MBH
Design OAT **95.0** °F
Equipment Sizing **(Auto-Sized) 39.2** MBH
Capacity Oversizing Factor **0** %
ARI Performance Rating **13.000** EER
DX System Configuration **2-stage compression, separate circuits**
Conventional Cutoff OAT **55.0** °F
Low Temperature Operation **Used**
Low Temperature Cutoff OAT **0.0** °F

Central Heating Unit - ASHP

Estimated Maximum Load **34.7** MBH
Design OAT **47.0** °F
Equipment Sizing **(Auto-Sized) 34.7** MBH
Capacity Oversizing Factor **0** %
ARI Performance Rating **4.000** COP
Cutoff OAT **-15.0** °F

Auxiliary Heating:

Auxiliary Heating Type **Electric Resistance**
Auxiliary Heating Upper Cutoff **40.0** °F

Proposed Input Data

Camp Lejeune - MRSG DST Storage
CRENSHAW CONSULTING ENGINEERS, INC.

11-29-2023
04:02PM

1. General Details:

Building Name **Proposed**

2. Plants Included in this Building:

None

3. Air Systems Included in this Building:

System Name	Mult.
[P] AHU	1

4. Miscellaneous Energy

(no items defined)

5: Meters

Electric **Camp Lejeune Electric Rate**
Natural Gas **Commerical Price (NC)**

6: Miscellaneous Data

Average Building Power Factor **100.00 %**
Source Electric Generating Efficiency **28.00 %**
Additional Floor Area **0.0 ft²**

II. Baseline Building HVAC Loads – Inputs

Space Input Data

Camp Lejeune - MRSRG DST Storage
CRENSHAW CONSULTING ENGINEERS, INC.

11-29-2023
04:02PM

[B] Restroom

1. General Details:

Floor Area 82.0 ft²
Avg. Ceiling Height 9.0 ft
Building Weight 70.0 lb/ft²

1.1. OA Ventilation Requirements:

Space Usage User-Defined
OA Requirement 1 0.0 CFM/person
OA Requirement 2 0.00 CFM/ft²
Space Usage Defaults ASHRAE Std 62.1-2010

2. Internals:

2.1. Overhead Lighting:

Fixture Type Recessed (Unvented)
Wattage 0.98 W/ft²
Ballast Multiplier 1.00
Schedule 90.1 Warehouse Lights/Elec

2.4. People:

Occupancy 0.0 Person
Activity Level Office Work
Sensible 245.0 BTU/hr/person
Latent 205.0 BTU/hr/person
Schedule None

2.2. Task Lighting:

Wattage 0.00 W/ft²
Schedule None

2.5. Miscellaneous Loads:

Sensible 0 BTU/hr
Schedule None
Latent 0 BTU/hr
Schedule None

2.3. Electrical Equipment:

Wattage 0.00 W/ft²
Schedule None

3. Walls, Windows, Doors:

Exp.	Wall Gross Area (ft ²)	Window 1 Qty.	Window 2 Qty.	Door 1 Qty.
W	90.0	0	0	0
S	90.0	0	0	1

3.1. Construction Types for Exposure W

Wall Type 2013 Zone 3 Above-Grade Steel-Framed Wall

3.2. Construction Types for Exposure S

Wall Type 2013 Zone 3 Above-Grade Steel-Framed Wall
Door Type 2013 Zone 3 Opaque Nonswinging

4. Roofs, Skylights:

(No Roof or Skylight data).

5. Infiltration:

Design Cooling 0.00 CFM
Design Heating 0.00 CFM
Energy Analysis 0.00 CFM
Infiltration occurs only when the fan is off.

6. Floors:

Type Floor Above Conditioned Space
(No additional input required for this floor type).

7. Partitions:

(No partition data).

Space Input Data

Camp Lejeune - MRSRG DST Storage
CRENSHAW CONSULTING ENGINEERS, INC.

11-29-2023
04:02PM

[B] Storage

1. General Details:

Floor Area **5440.0** ft²
Avg. Ceiling Height **10.0** ft
Building Weight **70.0** lb/ft²

1.1. OA Ventilation Requirements:

Space Usage **MISCELLANEOUS: Warehouse**
OA Requirement 1 **10.0** CFM/person
OA Requirement 2 **0.06** CFM/ft²
Space Usage Defaults **ASHRAE Std 62.1-2010**

2. Internals:

2.1. Overhead Lighting:

Fixture Type **Recessed (Unvented)**
Wattage **0.58** W/ft²
Ballast Multiplier **1.00**
Schedule **90.1 Warehouse Lights/Elec**

2.2. Task Lighting:

Wattage **0.00** W/ft²
Schedule **None**

2.3. Electrical Equipment:

Wattage **0.25** W/ft²
Schedule **90.1 Warehouse Lights/Elec**

3. Walls, Windows, Doors:

Exp.	Wall Gross Area (ft ²)	Window 1 Qty.	Window 2 Qty.	Door 1 Qty.
N	1584.0	0	0	0
S	1584.0	0	0	4
W	726.0	0	0	0
E	907.0	0	0	0

3.1. Construction Types for Exposure N

Wall Type **2013 Zone 3 Above-Grade Steel-Framed Wall**

3.2. Construction Types for Exposure S

Wall Type **2013 Zone 3 Above-Grade Steel-Framed Wall**
Door Type **2013 Zone 3 Opaque Nonswinging**

3.3. Construction Types for Exposure W

Wall Type **2013 Zone 3 Above-Grade Steel-Framed Wall**

3.4. Construction Types for Exposure E

Wall Type **2013 Zone 3 Above-Grade Steel-Framed Wall**

4. Roofs, Skylights:

(No Roof or Skylight data).

5. Infiltration:

Design Cooling **0.00** CFM
Design Heating **0.00** CFM
Energy Analysis **0.00** CFM
Infiltration occurs only when the fan is off.

6. Floors:

Type **Floor Above Conditioned Space**
(No additional input required for this floor type).

7. Partitions:

(No partition data).

2.4. People:

Occupancy **4.0** People
Activity Level **Medium Work**
Sensible **295.0** BTU/hr/person
Latent **455.0** BTU/hr/person
Schedule **90.1 Warehouse Occupants**

2.5. Miscellaneous Loads:

Sensible **0** BTU/hr
Schedule **None**
Latent **0** BTU/hr
Schedule **None**

Wall Constructions

Camp Lejeune - MRSRG DST Storage
CRENSHAW CONSULTING ENGINEERS, INC.

11-29-2023
04:03PM

2013 Zone 3 Above-Grade Steel-Framed Wall

Wall Details

Outside Surface Color **Medium**
Absorptivity **0.675**
Overall U-Value **0.077** BTU/(hr·ft²·°F)

Wall Layers Details (Inside to Outside)

Layers	Thickness in	Density lb/ft ³	Specific Ht. BTU / (lb·°F)	R-Value (hr·ft ² ·°F)/BTU	Weight lb/ft ²
Inside surface resistance	0.000	0.0	0.00	0.68000	0.0
Gypsum Board	0.625	50.0	0.26	0.56000	2.6
Board Insulation	0.720	2.0	0.22	5.00073	0.1
Batt Insulation + Steel Studs	3.500	50.0	0.16	6.00000	14.6
Gypsum Board	0.625	50.0	0.26	0.56000	2.6
Stucco	0.400	116.0	0.20	0.07994	3.9
Outside surface resistance	0.000	0.0	0.00	0.17000	0.0
Totals	5.870	-		13.05067	23.8

Window Constructions

Camp Lejeune - BEQ HP505
CRENSHAW CONSULTING ENGINEERS, INC.

11-06-2023
04:50PM

2013 Zone 3 Metal framing, operable

Window Details:

Detailed Input	No	
Height	4.00	ft
Width	1.00	ft
Overall U-Value	0.600	BTU/(hr·ft ² ·°F)
Overall Shade Coefficient	0.287	

Roof Constructions

Camp Lejeune - MRSB DST Storage
CRENSHAW CONSULTING ENGINEERS, INC.

11-29-2023
04:03PM

2013 Zone 2-3 Insulation Entirely Above Deck

Roof Details

Outside Surface Color **Medium**
Absorptivity **0.675**
Overall U-Value **0.039** BTU/(hr·ft²·°F)

Roof Layers Details (Inside to Outside)

Layers	Thickness in	Density lb/ft ³	Specific Ht. BTU / (lb·°F)	R-Value (hr·ft ² ·°F)/BTU	Weight lb/ft ²
Inside surface resistance	0.000	0.0	0.00	0.61000	0.0
22 gage steel deck	0.034	489.0	0.12	0.00011	1.4
Board Insulation	2.900	2.0	0.22	25.00000	0.5
Outside surface resistance	0.000	0.0	0.00	0.17000	0.0
Totals	2.934	-		25.78011	1.9

[B] Pkg CAV RTU w/Elec HP Input Data

Project Name: Camp Lejeune - MRSB DST Storage
Prepared by: CRENSHAW CONSULTING ENGINEERS, INC.

11-29-2023
04:03PM

1. General Details:

Air System Name [B] Pkg CAV RTU w/Elec HP
Equipment Type Packaged Rooftop Units
Air System Type Single Zone CAV
Number of zones 1

2. Ventilation System Components:

Ventilation Air Data:

Airflow Control Constant Ventilation Airflow
Ventilation Sizing Method ASHRAE Std 62.1-2010
Unocc. Damper Position Closed
Damper Leak Rate 0 %
Outdoor Air CO2 Level 400 ppm

Central Cooling Data:

Supply Air Temperature 55.0 °F
Coil Bypass Factor 0.100
Cooling Source Air-Cooled DX
Schedule JFMAMJJASOND
Capacity Control Cycled or Staged Capacity - Fan On

Central Heating Data:

Supply Temperature 95.0 °F
Heating Source Air Source Heat Pump
Schedule JFMAMJJASOND
Capacity Control Cycled or Staged Capacity - Fan On

Supply Fan Data:

Fan Type ASHRAE 90.1 App G Fan Curve
Configuration Draw-thru
Fan Performance ASHRAE Std 90.1-2010 Appendix G Fan kW
ASHRAE Std 90.1-2010 Baseline Fan Adjustment Factor Components:
Return and/or exhaust airflow control devices 0.50 in wg
Particulate Filtration Credit: MERV 13 through 15 0.90 in wg
Fan Control 1-speed fan, cooling and heating

Duct System Data:

Supply Duct Data:
Duct Heat Gain 0 %
Duct Leakage 0 %

Return Duct or Plenum Data:

Return Air Via Ducted Return

3. Zone Components:

Space Assignments:

Zone 1: Zone 1	
[B] Restroom	x1
[B] Storage	x1

Thermostats and Zone Data:

Zone	Cooling T-Stat Occ. (°F)	Cooling T-Stat Unocc. (°F)	Heating T-Stat Occ. (°F)	Heating T-Stat Unocc. (°F)	T-Stat Throttling Range (°F)	Diversity Factor (%)	Direct Exhaust Airflow (CFM)	Direct Exhaust Fan (kW)
1	75.0	85.0	70.0	60.0	1.50	100	0.0	0.0

Thermostat Schedule 90.1 Warehouse HVAC
Unoccupied Cooling is Available

Supply Terminals Data:

Zone	Terminal Type	Air Distribution	Air Distribution Effectiveness Specification	Air Distribution Effectiveness	Minimum Airflow	Fan Performance	Fan Efficiency	Design Supply Temp.
------	---------------	---------------------	---	--------------------------------------	-----------------	-----------------	-------------------	---------------------------

[B] Pkg CAV RTU w/Elec HP Input Data

Project Name: Camp Lejeune - MRSB DST Storage
Prepared by: CRENSHAW CONSULTING ENGINEERS, INC.

11-29-2023
04:03PM

1	Diffuser (no reheat)	Ceiling supply / ceiling return	ASHRAE 62.1	<auto>	0.00 CFM/person	-	-	-
---	----------------------	---------------------------------	-------------	--------	-----------------	---	---	---

Zone Heating Units:

Zone Unit Heat Source **Electric Resistance**
Zone Heating Unit Schedule **JFMAMJJASOND**

4. Sizing Data (Computer-Generated):

System Sizing Data:

Sizing Data:
Cooling Supply Temperature **55.0** °F
Supply Fan Airflow **802.7** CFM
Ventilation Airflow **428.2** CFM
Heating Supply Temperature **95.0** °F

Hydronic Sizing Specifications:

Chilled Water Delta-T **12.0** °F
Hot Water Delta-T **50.0** °F

Safety Factors:

Cooling Sensible **0** %
Cooling Latent **0** %
Heating **0** %

Zone Sizing Data:

Zone Airflow Sizing Method **Sum of space airflow rates**
Space Airflow Sizing Method **Individual peak space loads**

Zone	Supply Airflow (CFM)	Zone Htg Unit (MBH)	Reheat Coil (MBH)	- (CFM)
1	802.7	-	-	

5. Equipment Data

Central Cooling Unit - Air-Cooled DX

Estimated Maximum Load **49.6** MBH
Design OAT **95.0** °F
Equipment Sizing **(Auto-Sized) 57.1** MBH
Capacity Oversizing Factor **15** %
ASHRAE 90.1 Minimum Equipment Efficiency **(Auto-Calculated) 13.000** SEER
DX System Configuration **1-stage compression, 1 circuit**
Conventional Cutoff OAT **55.0** °F
Low Temperature Operation **Used**
Low Temperature Cutoff OAT **-15.0** °F

Central Heating Unit - ASHP

Estimated Maximum Load **38.0** MBH
Design OAT **47.0** °F
Equipment Sizing **(Auto-Sized) 47.5** MBH
Capacity Oversizing Factor **25** %
ASHRAE 90.1 Minimum Equipment Efficiency **(Auto-Calculated) 7.700** HSPF
Cutoff OAT **-15.0** °F

Auxiliary Heating:

Auxiliary Heating Type **Electric Resistance**
Auxiliary Heating Upper Cutoff **40.0** °F

Baseline Input Data

Camp Lejeune - MRSG DST Storage
CRENSHAW CONSULTING ENGINEERS, INC.

11-29-2023
04:03PM

1. General Details:

Building Name **Baseline**

2. Plants Included in this Building:

None

3. Air Systems Included in this Building:

System Name	Mult.
[B] Pkg CAV RTU w/Elec HP	1

4. Miscellaneous Energy

(no items defined)

5: Meters

Electric **Camp Lejeune Electric Rate**
Natural Gas **Commerical Price (NC)**

6: Miscellaneous Data

Average Building Power Factor **100.00 %**
Source Electric Generating Efficiency **28.00 %**
Additional Floor Area **0.0 ft²**

III. Proposed Building HVAC Loads – Outputs

Air System Sizing Summary for [P] AHU

Project Name: Camp Lejeune - MRSB DST Storage
Prepared by: CRENSHAW CONSULTING ENGINEERS, INC.

11-29-2023
04:05PM

Air System Information

Air System Name [P] AHU
Equipment Class SPLT AHU
Air System Type SZCAV

Number of zones 1
Floor Area 5522.0 ft²
Location Camp Lejeune, North Carolina

Sizing Calculation Information

Calculation Months Jan to Dec
Sizing Data Calculated

Zone CFM Sizing Sum of space airflow rates
Space CFM Sizing Individual peak space loads

Central Cooling Coil Sizing Data

Total coil load 3.2 Tons
Total coil load 38.4 MBH
Sensible coil load 29.6 MBH
Coil CFM at Aug 1500 5744 CFM
Max block CFM 5744 CFM
Sum of peak zone CFM 5744 CFM
Sensible heat ratio 0.772
CFM/Ton 1795.2
ft²/Ton 1725.9
BTU/(hr-ft²) 7.0
Water flow @ 10.0 °F rise N/A

Load occurs at Aug 1500
OA DB / WB 84.0 / 78.6 °F
Entering DB / WB 75.6 / 72.3 °F
Leaving DB / WB 70.8 / 70.5 °F
Coil ADP 70.3 °F
Bypass Factor 0.100
Resulting RH 86 %
Design supply temp. 58.0 °F
Zone T-stat Check 1 of 1 OK
Max zone temperature deviation 0.0 °F

Central Heating Coil Sizing Data

Max coil load 19.4 MBH
Coil CFM at Des Htg 5744 CFM
Max coil CFM 5744 CFM
Water flow @ 20.0 °F drop N/A

Load occurs at Des Htg
BTU/(hr-ft²) 3.5
Ent. DB / Lvg DB 66.6 / 69.7 °F

Supply Fan Sizing Data

Actual max CFM 5744 CFM
Standard CFM 5737 CFM
Actual max CFM/ft² 1.04 CFM/ft²

Fan motor BHP 4.25 BHP
Fan motor kW 3.37 kW
Fan static 2.50 in wg

Outdoor Ventilation Air Data

Design airflow CFM 374 CFM
CFM/ft² 0.07 CFM/ft²

CFM/person 93.51 CFM/person

Zone Sizing Summary for [P] AHU

Project Name: Camp Lejeune - MRSB DST Storage
Prepared by: CRENSHAW CONSULTING ENGINEERS, INC.

11-29-2023
04:05PM

Air System Information

Air System Name [P] AHU
Equipment Class SPLT AHU
Air System Type SZCAV

Number of zones 1
Floor Area 5522.0 ft²
Location Camp Lejeune, North Carolina

Sizing Calculation Information

Calculation Months Jan to Dec
Sizing Data Calculated

Zone CFM Sizing Sum of space airflow rates
Space CFM Sizing Individual peak space loads

Zone Terminal Sizing Data

Zone Name	Design Supply Airflow (CFM)	Minimum Supply Airflow (CFM)	Zone CFM/ft ²	Reheat Coil Load (MBH)	Reheat Coil Water gpm @ 20.0 °F	Zone Htg Unit Coil Load (MBH)	Zone Htg Unit Water gpm @ 20.0 °F	Mixing Box Fan Airflow (CFM)
Zone 1	5744	5744	1.04	0.0	-	0.0	-	0

Zone Peak Sensible Loads

Zone Name	Zone Cooling Sensible (MBH)	Time of Peak Sensible Cooling Load	Zone Heating Load (MBH)	Zone Floor Area (ft ²)
Zone 1	10.2	Jul 1600	13.3	5522.0

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (MBH)	Time of Peak Sensible Load	Air Flow (CFM)	Heating Load (MBH)	Floor Area (ft ²)	Space CFM/ft ²
Zone 1							
Restroom	1	0.3	Aug 1500	304	0.9	82.0	3.70
Storage	1	10.0	Jul 1600	5440	12.4	5440.0	1.00

Ventilation Sizing Summary for [P] AHU

Project Name: Camp Lejeune - MRSB DST Storage
 Prepared by: CRENSHAW CONSULTING ENGINEERS, INC.

11-29-2023
 04:05PM

1. Summary

Ventilation Sizing Method **ASHRAE Std 62.1-2010**
 Design Condition **Heating operation**
 Occupant Diversity (D) **1.000**
 Uncorrected Outdoor Air Intake (Vou) **366** CFM
 System Ventilation Efficiency (Ev) **0.980**
 Outdoor Air Intake (Vot) **374** CFM

2. Space Ventilation Analysis

		Supply Air (CFM)	Space Floor Area (ft²)	Area Outdoor Air Rate (CFM/ft²)	Time Averaged Occupancy (Occupants)	People Outdoor Air Rate (CFM/person)	Air Distribution Effectiveness	Space Outdoor Air (CFM)	Breathing Zone Outdoor Air (CFM)	Space Ventilation Efficiency
Zone Name / Space Name	Mult.	(Vpz)	(Az)	(Ra)	(Pz)	(Rp)	(Ez)	(Voz)	(Vbz)	(Evz)
Zone 1										
Restroom	1	304	82.0	0.00	0.0	0.00	0.8	0	0	1.064
Storage	1	5440	5440.0	0.06	4.0	10.00	0.8	458	366	0.980
Totals (incl. Space Multipliers)		5744							366	0.980

IV. Baseline Building HVAC Loads – Outputs

Air System Sizing Summary for [B] Pkg CAV RTU w/Elec HP

Project Name: Camp Lejeune - MRSB DST Storage
Prepared by: CRENSHAW CONSULTING ENGINEERS, INC.

11-29-2023
04:06PM

Air System Information

Air System Name [B] Pkg CAV RTU w/Elec HP
Equipment Class PKG ROOF
Air System Type SZCAV

Number of zones 1
Floor Area 5522.0 ft²
Location Camp Lejeune, North Carolina

Sizing Calculation Information

Calculation Months Jan to Dec
Sizing Data Calculated

Zone CFM Sizing Sum of space airflow rates
Space CFM Sizing Individual peak space loads

Central Cooling Coil Sizing Data

Total coil load 4.1 Tons
Total coil load 49.6 MBH
Sensible coil load 24.5 MBH
Coil CFM at Aug 1600 803 CFM
Max block CFM 803 CFM
Sum of peak zone CFM 803 CFM
Sensible heat ratio 0.495
CFM/Ton 194.1
ft²/Ton 1335.4
BTU/(hr-ft²) 9.0
Water flow @ 12.0 °F rise N/A

Load occurs at Aug 1600
OA DB / WB 83.4 / 78.5 °F
Entering DB / WB 80.2 / 71.5 °F
Leaving DB / WB 51.9 / 51.5 °F
Coil ADP 48.7 °F
Bypass Factor 0.100
Resulting RH 44 %
Design supply temp. 55.0 °F
Zone T-stat Check 1 of 1 OK
Max zone temperature deviation 0.0 °F

Central Heating Coil Sizing Data

Max coil load 38.0 MBH
Coil CFM at Des Htg 803 CFM
Max coil CFM 803 CFM
Water flow @ 50.0 °F drop N/A

Load occurs at Des Htg
BTU/(hr-ft²) 6.9
Ent. DB / Lvg DB 43.8 / 87.7 °F

Supply Fan Sizing Data

Actual max CFM 803 CFM
Standard CFM 802 CFM
Actual max CFM/ft² 0.15 CFM/ft²

Fan motor BHP n/a
Fan motor kW 0.89 kW

Outdoor Ventilation Air Data

Design airflow CFM 428 CFM
CFM/ft² 0.08 CFM/ft²

CFM/person 107.06 CFM/person

Zone Sizing Summary for [B] Pkg CAV RTU w/Elec HP

Project Name: Camp Lejeune - MRSB DST Storage
Prepared by: CRENSHAW CONSULTING ENGINEERS, INC.

11-29-2023
04:06PM

Air System Information

Air System Name [B] Pkg CAV RTU w/Elec HP
Equipment Class PKG ROOF
Air System Type SZCAV

Number of zones 1
Floor Area 5522.0 ft²
Location Camp Lejeune, North Carolina

Sizing Calculation Information

Calculation Months Jan to Dec
Sizing Data Calculated

Zone CFM Sizing Sum of space airflow rates
Space CFM Sizing Individual peak space loads

Zone Terminal Sizing Data

Zone Name	Design Supply Airflow (CFM)	Minimum Supply Airflow (CFM)	Zone CFM/ft ²	Reheat Coil Load (MBH)	Reheat Coil Water gpm @ 50.0 °F	Zone Htg Unit Coil Load (MBH)	Zone Htg Unit Water gpm @ 50.0 °F	Mixing Box Fan Airflow (CFM)
Zone 1	803	803	0.15	0.0	-	0.0	-	0

Zone Peak Sensible Loads

Zone Name	Zone Cooling Sensible (MBH)	Time of Peak Sensible Cooling Load	Zone Heating Load (MBH)	Zone Floor Area (ft ²)
Zone 1	16.8	Aug 1700	20.5	5522.0

Space Loads and Airflows

Zone Name / Space Name	Mult.	Cooling Sensible (MBH)	Time of Peak Sensible Load	Air Flow (CFM)	Heating Load (MBH)	Floor Area (ft ²)	Space CFM/ft ²
Zone 1							
[B] Restroom	1	0.4	Aug 1700	40	1.1	82.0	0.49
[B] Storage	1	16.4	Aug 1700	762	19.5	5440.0	0.14

Ventilation Sizing Summary for [B] Pkg CAV RTU w/Elec HP

Project Name: Camp Lejeune - MRSB DST Storage
 Prepared by: CRENSHAW CONSULTING ENGINEERS, INC.

11-29-2023
 04:06PM

1. Summary

Ventilation Sizing Method **ASHRAE Std 62.1-2010**
 Design Condition **Heating operation**
 Occupant Diversity (D) **1.000**
 Uncorrected Outdoor Air Intake (Vou) **366** CFM
 System Ventilation Efficiency (Ev) **0.856**
 Outdoor Air Intake (Vot) **428** CFM

2. Space Ventilation Analysis

		Supply Air (CFM)	Space Floor Area (ft²)	Area Outdoor Air Rate (CFM/ft²)	Time Averaged Occupancy (Occupants)	People Outdoor Air Rate (CFM/person)	Air Distribution Effectiveness	Space Outdoor Air (CFM)	Breathing Zone Outdoor Air (CFM)	Space Ventilation Efficiency
Zone Name / Space Name	Mult.	(Vpz)	(Az)	(Ra)	(Pz)	(Rp)	(Ez)	(Voz)	(Vbz)	(Evz)
Zone 1										
[B] Restroom	1	40	82.0	0.00	0.0	0.00	0.8	0	0	1.456
[B] Storage	1	762	5440.0	0.06	4.0	10.00	0.8	458	366	0.856
Totals (incl. Space Multipliers)		803							366	0.856

V. Energy Model Inputs

Design Weather Parameters & MSHGs

Camp Lejeune - MRSRG DST Storage
CRENSHAW CONSULTING ENGINEERS, INC.

11-29-2023
04:06PM

Design Parameters:

City Name **Camp Lejeune**
 Location **North Carolina**
 Latitude **34.3** Deg.
 Longitude **77.9** Deg.
 Elevation **33.0** ft
 Summer Design Dry-Bulb **84.0** °F
 Summer Coincident Wet-Bulb **78.6** °F
 Summer Daily Range **19.0** °F
 Winter Design Dry-Bulb **22.0** °F
 Winter Design Wet-Bulb **19.2** °F
 Atmospheric Clearness Number **0.95**
 Average Ground Reflectance **0.20**
 Soil Conductivity **0.800** BTU/(hr·ft·°F)
 Local Time Zone (GMT +/- N hours) **5.0** hours
 Consider Daylight Savings Time **No**
 Simulation Weather Data **(EXT)**
 Current Data is **User Modified**
 Design Cooling Months **January to December**

Design Day Maximum Solar Heat Gains

(The MSHG values are expressed in BTU/(hr·ft²))

Month	N	NNE	NE	ENE	E	ESE	SE	SSE	S
January	21.3	21.3	23.5	94.6	159.4	212.6	233.8	239.0	237.4
February	25.1	25.1	55.6	136.5	191.1	227.7	236.5	225.1	215.9
March	29.2	31.9	102.9	167.5	214.0	227.4	218.6	192.0	175.7
April	33.2	69.8	141.4	187.8	211.8	210.8	180.7	141.8	119.6
May	36.0	99.0	161.7	198.6	205.6	192.0	150.9	102.4	79.4
June	46.3	111.1	167.9	199.1	201.0	182.6	136.9	86.8	65.5
July	36.9	101.5	159.5	192.8	203.1	187.8	147.3	100.0	77.5
August	34.8	71.3	136.8	178.9	205.9	203.0	175.1	137.2	115.7
September	30.2	30.2	99.4	154.3	201.9	219.0	209.2	185.6	169.8
October	25.7	25.7	59.5	124.8	186.8	219.4	229.1	218.5	208.4
November	21.4	21.4	21.4	93.5	159.0	205.4	233.2	237.2	232.1
December	19.5	19.5	19.5	79.0	143.0	201.6	228.2	240.7	240.2
Month	SSW	SW	WSW	W	WNW	NW	NNW	HOR	Mult
January	239.5	236.6	211.1	162.9	86.7	25.3	21.3	156.0	1.00
February	224.1	234.9	229.1	187.6	135.8	58.8	25.1	196.4	1.00
March	190.8	217.7	228.0	211.3	170.7	102.0	31.7	231.7	1.00
April	141.2	181.7	208.8	213.1	190.5	139.0	72.1	251.8	1.00
May	102.6	152.6	189.8	209.1	199.1	159.4	101.7	259.3	1.00
June	87.1	139.1	180.0	204.9	199.3	164.1	113.1	260.0	1.00
July	100.1	148.9	185.1	205.2	194.7	155.4	102.6	256.6	1.00
August	136.5	175.4	201.1	205.7	184.2	135.2	71.5	247.1	1.00
September	185.0	208.2	219.4	200.3	157.4	99.6	30.2	222.5	1.00
October	218.9	229.6	216.9	187.1	128.0	57.4	25.7	191.1	1.00
November	237.1	232.9	206.3	158.0	94.1	21.4	21.4	153.2	1.00
December	241.7	232.3	197.9	147.3	77.2	19.5	19.5	137.5	1.00

Mult. = User-defined solar multiplier factor.

Simulation Weather Summary for New River Mcaf, NC (EXT)

Camp Lejeune - MRSB DST Storage
CRENSHAW CONSULTING ENGINEERS, INC.

11-29-2023
04:06PM

Table 1. Descriptive Parameters:

City	New River Mcaf
Location	NC
Type of Data	(EXT)
Latitude	34.7 Deg.
Longitude	77.4 Deg.
Elevation	16.4 ft
Local Time Zone (GMT +/- N hours)	5.0 hours
Average Ground Reflectance	0.20

Table 2. Dry-Bulb Temperature Statistics (°F):

Month	Absolute Maximum	Average Maximum	Average	Average Minimum	Absolute Minimum
January	75.2	52.0	42.3	32.9	19.4
February	79.0	56.6	45.8	34.5	9.0
March	77.0	59.6	49.0	39.8	21.9
April	82.4	72.5	61.8	51.6	30.2
May	91.4	78.0	69.2	60.5	48.2
June	95.0	85.0	76.4	69.0	57.0
July	93.2	84.5	76.3	68.7	59.0
August	99.0	88.2	79.5	72.0	62.6
September	107.8	85.4	77.2	70.3	63.9
October	84.2	73.8	64.6	56.3	44.6
November	79.7	71.2	58.6	47.0	28.9
December	75.0	55.4	43.5	32.8	21.9

Table 3. Daily Solar Radiation Statistics:

Month	Daily Total Solar on Horizontal (BTU/ft²)			Daily Clearness Number (dimensionless)		
	Maximum	Average	Minimum	Maximum	Average	Minimum
January	1148.5	758.2	189.5	0.705	0.462	0.111
February	1362.5	888.4	247.4	0.602	0.426	0.133
March	1948.9	1192.8	381.4	0.697	0.449	0.142
April	2405.0	1837.3	493.6	0.710	0.577	0.157
May	2490.0	1621.9	631.2	0.687	0.459	0.179
June	2365.4	1356.3	753.6	0.644	0.370	0.206
July	2375.2	1615.9	614.2	0.653	0.450	0.175
August	2224.4	1574.3	372.4	0.657	0.476	0.119
September	2015.6	1406.9	323.4	0.653	0.495	0.113
October	1671.2	1133.4	356.0	0.663	0.499	0.175
November	1295.2	991.9	372.3	0.673	0.564	0.228
December	1056.5	706.5	226.0	0.672	0.469	0.150

Simulation Weather Summary for New River Mcaf, NC (EXT)

Camp Lejeune - MRSO DST Storage
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11-29-2023
04:06PM

Table 4. Time of Occurrence for Maximums and Minimums:

Month	Highest Dry-Bulb Temperature	Lowest Dry-Bulb Temperature	Maximum Total Solar	Minimum Total Solar
January	Jan 3, 1300	Jan 27, 2300	Jan 16	Jan 23
February	Feb 26, 1400	Feb 5, 0600	Feb 24	Feb 2
March	Mar 15, 1300	Mar 9, 0500	Mar 23	Mar 17
April	Apr 23, 1100	Apr 6, 0400	Apr 30	Apr 12
May	May 10, 1300	May 5, 0400	May 30	May 15
June	Jun 8, 1300	Jun 14, 0400	Jun 14	Jun 5
July	Jul 11, 1300	Jul 14, 0400	Jul 7	Jul 27
August	Aug 1, 1400	Aug 31, 0400	Aug 10	Aug 30
September	Sep 3, 1100	Sep 10, 0200	Sep 1	Sep 14
October	Oct 2, 1300	Oct 12, 0500	Oct 3	Oct 29
November	Nov 28, 1200	Nov 22, 0600	Nov 4	Nov 25
December	Dec 3, 1100	Dec 25, 0600	Dec 1	Dec 31

Table 5. Calendar Data:

Day of Week for January 1st **Wednesday**

Holidays:

No holidays have been defined.

Electric Rate Input Data

Camp Lejeune - MRSO DST Storage
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11-29-2023
04:07PM

Camp Lejeune Electric Rate

General Details

Rate Name	Camp Lejeune Electric Rate	
Rate Type	Simple	
Energy Units	kWh	
Conversion	1.00000	kWh/kWh
Demand Units	kW	
Flat Price	0.12240	\$/kWh
Customer Charge	0.00	\$
Minimum Charge	0.00	\$
Tax Rate	0.00	%

Commerical Price (NC)

General Details

Rate Name Commerical Price (NC)
Rate Type Simple
Energy Units Therm
Conversion 100.00000 kBTU/Therm
Demand Units Hourly Peak
Flat Price 1.06800 \$/Therm
Customer Charge 0.00 \$
Minimum Charge 0.00 \$
Tax Rate 0.00 %

VI. Proposed vs Baseline Energy Model Outputs

LEED v4 EA Credit Optimize Energy Performance Summary Report

Camp Lejeune - MRSG DST Storage
CRENSHAW CONSULTING ENGINEERS, INC.

11-29-2023
04:07PM

1. REPORT AND PROJECT INFORMATION

General

Simulation Program Name and Version Hourly Analysis Program 5.11
Simulation Weather File Name New River Mcaf, NC (EXT)

Building Designations

Proposed Building Proposed
Baseline - 0° Baseline
Baseline - 90° n/a
Baseline - 180° n/a
Baseline - 270° n/a

Space Summary

	Proposed Design	Baseline
Conditioned Floor Area (ft²)	5,522	5,522
Unconditioned Floor Area (ft²)	0	0
Total Floor Area (ft²)	5,522	5,522

2. MINIMUM ENERGY PERFORMANCE CALCULATOR

SHADING AND FENESTRATION

Building Massing and Zoning

Above-Grade Wall & Vertical Glazing Areas

Orientation	Baseline Design (0° rotation)			Proposed Design		
	Gross Above-Grade Wall Area	Vertical Glazing Area		Gross Above-Grade Wall Area	Vertical Glazing Area	
	(ft²)	(ft²)	(% WWR)	(ft²)	(ft²)	(% WWR)
North	1,584	0	0.0	1,584	0	0.0
North-Northeast	0	0	0.0	0	0	0.0
Northeast	0	0	0.0	0	0	0.0
East-Northeast	0	0	0.0	0	0	0.0
East	907	0	0.0	907	0	0.0
East-Southeast	0	0	0.0	0	0	0.0
Southeast	0	0	0.0	0	0	0.0
South-Southeast	0	0	0.0	0	0	0.0
South	1,674	0	0.0	1,674	0	0.0
South-Southwest	0	0	0.0	0	0	0.0
Southwest	0	0	0.0	0	0	0.0
West-Southwest	0	0	0.0	0	0	0.0
West	816	0	0.0	816	0	0.0
West-Northwest	0	0	0.0	0	0	0.0
Northwest	0	0	0.0	0	0	0.0
North-Northwest	0	0	0.0	0	0	0.0
Total	4,981	0	0.0	4,981	0	0.0

Roof & Skylight Areas

Baseline Design (0° rotation)			Proposed Design		
Gross Roof Area	Skylight Area		Gross Roof Area	Skylight Area	
(ft²)	(ft²)	(%)	(ft²)	(ft²)	(%)
0	0	0.0	0	0	0.0

Note: In these tables, roof and skylight surfaces with slope of 60° or more (from horizontal) are treated as walls and vertical glazing, as according to ASHRAE 90.1 Section 3.

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Camp Lejeune - MRSO DST Storage
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11-29-2023
04:07PM

PERFORMANCE OUTPUTS

Energy Sources

Energy Type	Energy Consumption Units	Demand Units	Utility Rate Name	Utility Rate Structure
Electric	kWh	kW	Camp Lejeune Electric Rate	
Natural Gas	Therm	MBH	Commerical Price (NC)	

Performance Rating Method Compliance

Baseline Energy Summary by End Use

End Use	Unregulated ?	Baseline Design Energy Type	Units of Annual Energy & Peak Demand	Baseline (0° rotation)	Baseline (90° rotation)	Baseline (180° rotation)	Baseline (270° rotation)	Baseline Design Total (Average of 4 Rotations)
Interior Lighting		Electric	Consumption [kWh]	8,231.7	0.0	0.0	0.0	8,231.7
			Demand [kW]	2.9	0.0	0.0	0.0	2.9
Space Heating		Electric	Consumption [kWh]	1,586.7	0.0	0.0	0.0	1,586.7
			Demand [kW]	8.7	0.0	0.0	0.0	8.7
Space Cooling		Electric	Consumption [kWh]	2,906.5	0.0	0.0	0.0	2,906.5
			Demand [kW]	3.8	0.0	0.0	0.0	3.8
Pumps		Electric	Consumption [kWh]	0.0	0.0	0.0	0.0	0.0
			Demand [kW]	0.0	0.0	0.0	0.0	0.0
Heat Rejection		Electric	Consumption [kWh]	0.0	0.0	0.0	0.0	0.0
			Demand [kW]	0.0	0.0	0.0	0.0	0.0
Fans - Interior		Electric	Consumption [kWh]	2,956.1	0.0	0.0	0.0	2,956.1
			Demand [kW]	0.9	0.0	0.0	0.0	0.9
Receptacle Equipment	X	Electric	Consumption [kWh]	3,460.0	0.0	0.0	0.0	3,460.0
			Demand [kW]	1.2	0.0	0.0	0.0	1.2
Total energy consumption by energy type		Electric	kWh	19,141.1	0.0	0.0	0.0	19,141.1
		Natural Gas	Therm	0.0	0.0	0.0	0.0	0.0

(1) This form determines compliance using cost calculations from Section 1.9. Process Energy Costs should be modeled to accurately reflect the proposed building. Process Energy must be the same in the baseline and proposed cases, unless an exceptional calculation is used. Process energy costs must be at least 25% of the total baseline energy costs. Any exceptions must be supported by a narrative and/or other supporting documentation.

(2) In this project Process Energy is 18.1% of total baseline energy cost.

Baseline Building Annual Energy Cost by Energy Type

Energy Type	Baseline (0° rotation) (\$)	Baseline (90° rotation) (\$)	Baseline (180° rotation) (\$)	Baseline (270° rotation) (\$)	Baseline Design Total (\$)
Electric kWh	2,343	0	0	0	2,343
Natural Gas Therm	0	0	0	0	0
Baseline Annual Energy Cost	2,343	0	0	0	2,343

LEED v4 EA Credit Optimize Energy Performance Summary Report

Camp Lejeune - MRSB DST Storage
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11-29-2023
04:07PM

Proposed Energy Summary by End Use

End Use	Unregulated ?	Energy Type	Units of Annual Energy and Peak Demand	Baseline	Proposed	Energy / Demand Savings per End Use	End Use Percent Contribution to Total Energy Savings	End Use Percent Contribution to Total Cost Savings	Percent of Total Proposed Site Energy Consumption
Interior Lighting		Electric	Consumption [kWh]	8,232	3,084	62.5 %	104.6 %	104.6 %	21.7 %
			Demand [kW]	2.9	1.1	62.5 %			
Space Heating		Electric	Consumption [kWh]	1,587	1,010	36.3 %	11.7 %	11.7 %	7.1 %
			Demand [kW]	8.7	11.0	-26.3 %			
Space Cooling		Electric	Consumption [kWh]	2,907	989	66.0 %	39.0 %	39.0 %	7.0 %
			Demand [kW]	3.8	1.4	64.1 %			
Pumps		Electric	Consumption [kWh]	0	0	n/a	0.0 %	0.0 %	0.0 %
			Demand [kW]	0.0	0.0	n/a			
Heat Rejection		Electric	Consumption [kWh]	0	0	n/a	0.0 %	0.0 %	0.0 %
			Demand [kW]	0.0	0.0	n/a			
Fans - Interior		Electric	Consumption [kWh]	2,956	5,676	-92.0 %	-55.3 %	-55.3 %	39.9 %
			Demand [kW]	0.9	3.4	-280.6 %			
Receptacle Equipment	X	Electric	Consumption [kWh]	3,460	3,460	0.0 %	0.0 %	0.0 %	24.3 %
			Demand [kW]	1.2	1.2	0.0 %			

Performance Rating Energy Consumption and Cost by Fuel Type - Performance Rating Method Compliance

Energy Type	Site Energy Units	Baseline		Proposed		Percent Savings	
		Site Energy Use (Units shown per energy type)	Cost(\$)	Site Energy Use (Units shown per energy type)	Cost (\$)	Site Energy Use	Cost
Electric	kWh	19,141.1	2,343	14,218.7	1,740	25.7%	25.7%
Natural Gas	Therm	0.0	0	0.0	0	n/a	n/a
Energy Model Subtotal	kWh	19,141.1	2,343	14,218.7	1,740	25.7%	25.7%

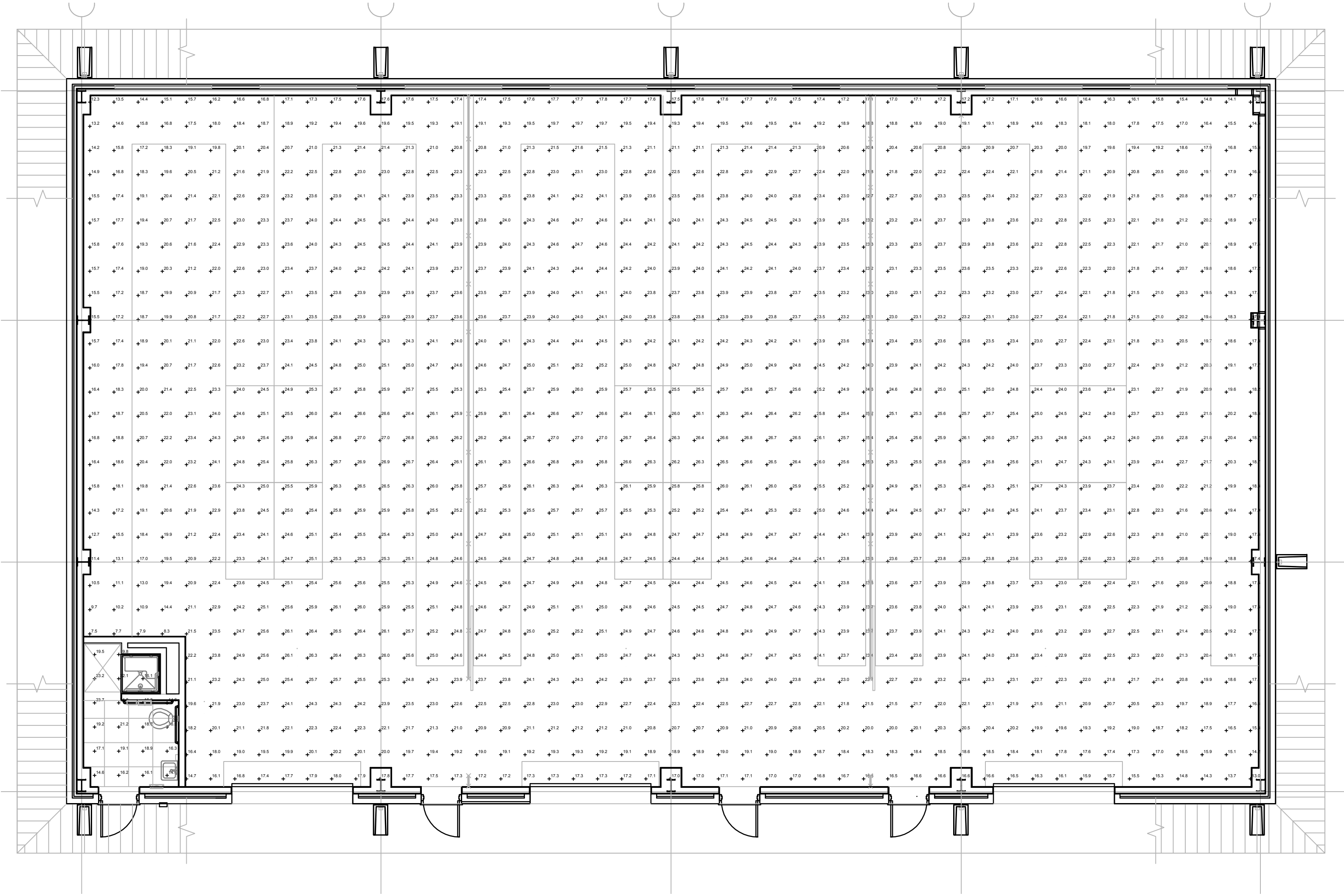
Unmet Loads

Unmet Loads	Baseline Building (0° rotation)	Proposed Building	Difference
Number of hours heating loads not met	85	6	+79
Number of hours cooling loads not met	18	0	+18
Totals	103	6	+97
Compliance	Yes		

Appendix C – ELECTRICAL DESIGN ANALYSIS

**CAMP LEJEUNE – MRSG DST STORAGE
FACILITY**

**ELECTRICAL DESIGN ANALYSIS
100% SUBMITTAL**



Statistics					
Description	Avg	Max	Min	Max/Min	Avg/Min
STORAGE	22.2 fc	27.0 fc	7.5 fc	3.6:1	3.0:1
BATHROOM	18.4 fc	23.7 fc	14.6 fc	1.6:1	1.3:1